Martin's Life Work

October 29, 2026

Chapter 1

Calculus

We start with Real Analysis. This loosely following [2]

1.1 Set Theory

The elements of a set can be chosen abstractly. For instance, in a university, a set of people can contain students, teachers, and faculty. We denote the objects of the set as generalized things. For instance, numbers, animals, and personal emotions are all things that we can place in a set.

Denote the set by the curly-brackets $\{\}$. The following operations all

Definition 1.1.

- *Let $\cup$ define set union. Ie, if $x \in A, y \in B, \{x, y\} \in A \cup B$.*
- *Let $\cap$ define set intersection. This is defined as if $x, y \in A \cap B$, then $x \in A, x \in B, y \in A, y \in B$.*
- *Let A be a set. Let B be a subset of A, that is, $B \subset A$. Then, let B^c denote the compliment of B, or the points in A that are not in B. If $A = B$, then $B^c = \emptyset$.*

We can use these operations to discuss some interesting groups of numbers. The following are a few useful sets.

Definition 1.2. *The following are commonly used sets;*

- *Natural numbers:* $\mathbb{N} = \{1, 2, 3, \dots\}$
- *Whole Numbers:* $\mathbb{W} = \{0, 1, 2, 3, \dots\}$
- *Integers:*$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$
- *Rational Numbers:* $\mathbb{Q} = \left\{ \frac{x}{y} : \{x, y\} \in \mathbb{Z}, y \neq 0 \right\}$
- *Irrational Numbers:* $\mathbb{P} = \{x : x \text{ cannot be written as a fraction.}\}$
- *Real Numbers:* $\mathbb{R} = \mathbb{Q} \cup \mathbb{P} = \mathbb{Q} \cup \mathbb{Q}^c$

The splitting property for the overlap of two sets is defined by a useful identity. Let A, B be two non-empty, non-disjoint sets. The set B can be written as;

$$B = (B \cap A) \cup (B \cap A^c) . \tag{1.1}$$

this is illustrated in Fig 1.1.

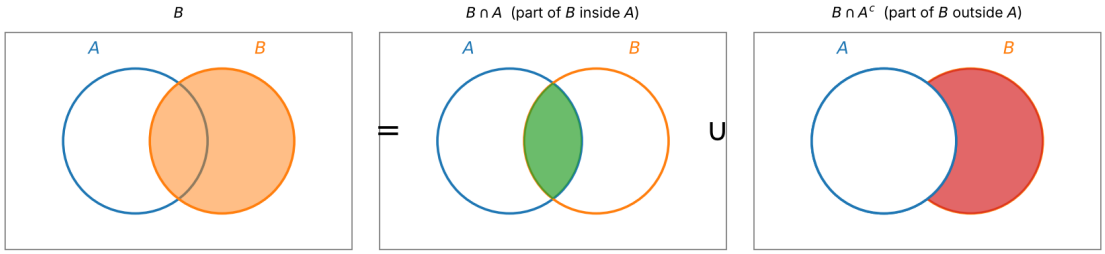


Figure 1.1: Splitting identity of two sets.

1.1.1 Properties of Set Operations

The operations of set union and intersection allow us to learn about the properties of sets under union, intersection, and compliment. We can define a few interesting theorems or results that allow us to start getting some logic in the world.

Theorem 1.1.

$$X^c \cap Y^c = (X \cup Y)^c \tag{1.2}$$

Proof. *To prove this, we need to show that the LHS is equal to the RHS. Consider $X \cup Y$. If $x \in X^c$ then $x \notin X$. If $y \in Y^c$, $y \notin Y$. The overlap of the compliment sets, $X^c \cap Y^c$, contains neither x nor y. Thus, $x, y \in X \cup Y$, This completes the proof as since $x, y \notin X^c \cap Y^c$, then we must have that $(X \cup Y)^c = X^c \cap Y^c$.* ■

For a union or intersection of many sets, define the notation;

$$\bigcup_{i=1}^{\infty} X_i = X_1 \cup X_2 \cup \ldots, \quad \bigcap_{i=1}^{\infty} X_i = X_1 \cap X_2 \cap \ldots$$

Theorem 1.2.

$$\left(\bigcap_{i=1}^{\infty} X_i \right)^c = \bigcup_{i=1}^{\infty} X_i^c \tag{1.3}$$

Proof. *Let $x_i \in X_i$, $x_i \notin X_i^c$. Split this into two cases, where $\cap X_i \neq \emptyset$ and $\cap X_i = \emptyset$.*

- *$\cap X_i = \emptyset$, then for every x_i there exists a disjoint set X_j which doesn't contain x_i. This implies that $x_i \in X_j^c$.*
- *$\cap X_i \neq \emptyset$. If the intersection is not empty, then there exists $x_i \in X_1, X_2, \ldots$. Thus, since x_i is simultaneously contained within every intersected set; $x_i \notin \cup X_i^c$.*

This completes the proof stating that the compliment of the intersection of all sets will contain all points outside of the intersection which is the union of each individual compliment. ■

1.1.2 Functions

We can furthermore define functions as sets by adressing rules. Depending on these rules, a function can map an element in set A to set B. This is defined using the following notation;

$$f : A \to B \tag{1.4}$$

Example 1.1. *Suppose $f(x) = 3x^2$ for $x \in \mathbb{R}$. Define the set $\mathbb{R}_+ = [0, \infty)$. Thus, we would say that;*

$$f : \mathbb{R} \to \mathbb{R}_+ \tag{1.5}$$

We can classify functions based on the uniqueness of their mappings. For a function $f : A \to B$, we assume that each element in A maps to at most one element in B. A general function takes an element in A and maps to at most one value in B. Not all elements need to be mapped to, and some points can be visited multiple times. **Injectivity** implies that each input on A produces a unique result in B. Ie, there are no duplicates. However, not all elements in B are nessecarily covered. **Surjectivity** implies that all outputs in B are mapped to, however, multiple $x \in A$ may map to elements in B. A map is **bijective** if every point in A uniquely maps to a point in B. This is also known as a set which is injective and surjective. See Fig. 1.2 for a visualization.

There are some useful notation. Here, for $f : X \to Y$, define the **image** as the set $Y = f(X)$. similarily, define the inverse image, or **preimage** as the set $X = f^{-1}(Y)$.

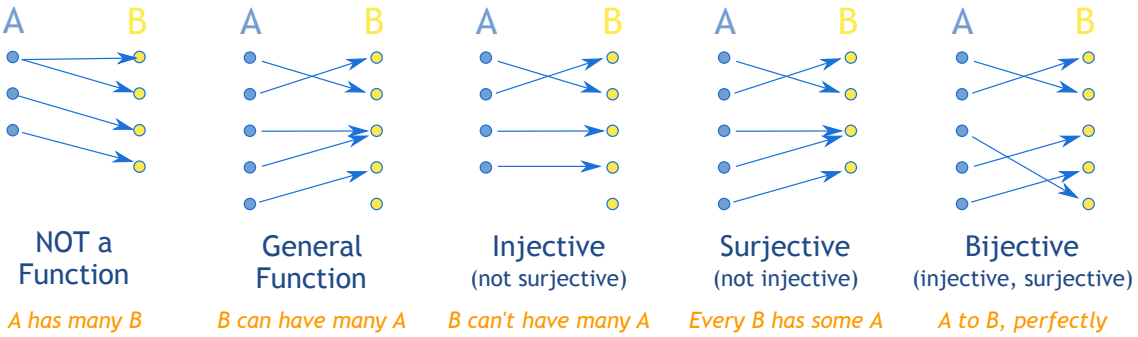


Figure 1.2: Function types based on correspondance. For a function $f : A \to B$, we assume that each element in A maps to at most one element in B. A general function takes an element in A and maps to at most one value in B. Not all elements need to be mapped to, and some points can be visited multiple times. **Injectivity** implies that each input on A produces a unique result in B. Ie, there are no duplicates. However, not all elements in B are nessecarily covered. **Surjectivity** implies that all outputs in B are mapped to, however, multiple $x \in A$ may map to elements in B. A map is **bijective** if every point in A uniquely maps to a point in B.

Definition 1.3. *general functions have the following properties when used in tandem with set union and intersection. In addition, we present the inverse operations.*

a. $f(A \cup B) = f(A) \cup f(B)$

b. $f(A \cap B) \subseteq f(A) \cap f(B)$

c. $f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$

d. $f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$

e. $f^{-1}(f(A)) \supseteq A$

f. $f(f^{-1}(C)) \subseteq C$

Proof. We aim to prove each statement mentioned above. Let $f : X \to Y$ and $A, B \subset X$, $C \subset Y$.

a. Let $x \in f(A \cup B)$. Thus, there exists a y such that $y \in A \cup B \to x \in f(y)$ Since $y \in A, \to x \in f(A)$ or $y \in B \to x \in f(B)$. Thus, $f(A \cup B) \subseteq f(A) \cup f(B)$. Similarily, in the converse, suppose that $x \in f(A)$. then, there will exists a $y \in A$. Since $y \in A, y \in (A \cup B)$. Thus, $f(A) \subseteq f(A \cup B)$. Or more relevantly, $f(A) \cup f(B) \subseteq f(A \cup B)$. The converse with B is trivial. Thus, $f(A \cup B) = f(A) \cup f(B)$.

b. Let $x \in f(A \cap B)$. There exists a y such that $y \in A$ and $y \in B$, so $y \in A \cap B$. So, if $y \in A$, $x \in f(A)$ and $y \in B$, $x \in f(B)$. If both are true simultaneously, then $f(A \cap B) \subseteq f(A) \cap f(B)$.

c. Assume that $x \in f^{-1}(A \cup B)$ Then, $f(x) \in A \cup B$. Since $f(x) \in A$ and $f(x) \in B$ and for all A, B there exists a preimage, then $f^{-1}(A \cup B) \subseteq f(A) \cup f(B)$. The second inclusion is shown

by the converse. Let $x \in f^{-1}(A) \cup f^{-1}(B)$. Since $f(x) \in A \cup B$ then $f^{-1}(A) \cup f^{-1}(B) \subseteq f^{-1}(A \cup B)$. Both inclusions are true, so there is an equality.

d. Let $y \in f^{-1}(A \cap B)$ such that $f(y) \in A \cap B$. So, $f(y) \in A$ and $f(y) \in B$. Thus, $y \in f^{-1}(A)$ and $y \in f^{-1}(B)$. Or $y \in f^{-1}(A) \cap f^{-1}(B)$. So, $f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$.

e. Let $x \in f^{-1}(f(A))$. The inverse image of $f(A)$ will point to A if f is injective. Otherwise, it is at most a subset of A. $f^{-1}(f(A)) \supseteq A$

f. Let $x \in f(f^{-1}(A))$. The image of the preimage is the identity if every element in A is mapped to. Thus, the identity is equal if f is surjective.

■

Remark that if f is a bijective map, then $f^{-1}(f(A)) = f(f^{-1}(A))$.

1.1.3 Problems

Problem 1.1. *Let $\leq, \geq$ have their regular definitions. Let $A = \{0 \leq x \leq 3, x \in \mathbb{R}\}, B = \{-1 \leq x \leq 1, x \in \mathbb{R}\}$. Show that $A \cap B = C = \{0 \leq x \leq 1, x \in \mathbb{R}\}$*

Solution. To show this, we need to show that the LHS is equal to the RHS for the final line in the problem statement. Trivially, $C \subset A$; $\inf C \in A, \sup C \in A$. Thus, $\forall x \in C, x \in A$. Similarily, $C \subset B$ by the same logic. Thus, state $C^c = (-\infty, 0) \cup (1, \infty)$. Furthermore, writing the union of elements in the compliments of A, B; $A^c \cup B^c = (-\infty, 0) \cup (1, \infty)$. We use the property of intersections of sets

$$\begin{aligned} A^c \cup B^c &= (A \cap B)^c \\ C^c &= (A \cap B)^c \\ C &= A \cap B \end{aligned}$$

■

Problem 1.2. *List elements of;*

$$\big(\{2,3,4\} \cup \{x \in \mathbb{R} : x^2 - 4x + 3 = 0\}\big) \cap \{x \in \mathbb{R} : -1 \leq x < 3\} \tag{1.6}$$

Solution. The LHS of this contains two sets. The right-set are solutions to a quadratic equation. This contains two elements.

$$x_{1,2} = \frac{4 \pm \sqrt{16 - 12}}{2} = \{1, 3\}$$

Thus, the LHS is the union of two sets. Rewriting this;

$$\{1,2,3,4\} \cap \{x \in \mathbb{R} : -1 \leq x < 3\} \tag{1.7}$$

The intersection of the discrete set of integers between 1 and 4 and the open neighborhood between -1 and 3 is simply $\{1, 2\}$ as 3 is not included in the set.

$$\big(\{2,3,4\} \cup \{x \in \mathbb{R} : x^2 - 4x + 3 = 0\}\big) \cap \{x \in \mathbb{R} : -1 \leq x < 3\} = \{1,2\} \tag{1.8}$$

Problem 1.3. *If $A \subset S$, show the following;*

a. $(A^c)^c = A$

b. $A \cup A = A \cap A = A \cup \emptyset = A$

c. $A \cap \emptyset = \emptyset$

d. $A \times \emptyset = \emptyset$

Solution.

1. Let $x \in A^c$. $x \notin A$. Thus, $(A^c)^c = A$.
2. If $x \in A$ and $x \in A$ and A is the same set, logically, the union of both sets is just the set. Similarily, the intersection with itself is just itself as all points are covered witin the same set. The set $\emptyset$ is defined as the set which contains nothing. Thus, $A \cup \emptyset$ is the union of the set A with nothing. Since $\emptyset$ doesn't add anything new to the set, one logically writes that $A \cup \emptyset = A$.
3. let $x \in A$ Since $x \notin \emptyset$ then $x \notin A \cap \emptyset$. However, nothing intersects with the emptyset, so naturally, $A \cap \emptyset = \emptyset$ as there is nothing to intersect over.
4. The cartesian product of the emptyset is again defined by the property of the emptyset having no elements. Since there is no ordered pair formed when taking the cartesian product with the emptyset, the logical output set is a set with no elements, which is the emptyset.

Problem 1.4. *Prove the following statements:*

1. $A^c \cup B^c = (A \cap B)^c$
2. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
3. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Problem 1.5. *We prove each statement from top to bottom. To show equalities, we need to show that the LHS equates the the RHS.*

Let $x \in A \cap B$. Since x is in both A and B, then $x \notin A^c \cup B^c$. Thus, since x is general; $A^c \cup B^c = (A \cap B)^c$. Similarily, let $x \in A^c \cup B^c$, then by logical, x can be every value within except the points which overlap in A and B as their complements do not touch points which are contained in both sets.

Let $x \in A \cap (B \cup C)$ This set is in both A and $(B \cup C)$. Thus, $x \in A, B \cup C \rightarrow, A, B \rightarrow, x \in (A \cap B)$. $x \in A, B \cup C \rightarrow, A, C \rightarrow, x \in (A \cap C)$. So, x must be in one or the other. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Let $x \in A \cup (B \cap C)$, or x is in A or B and C. Thus, we have two inclusions, $x \in A \cup B$, or its in A or B and $A \cup C$, or A or C. Since x must at least be in A, the left term is easy to see, however, if $x \in (B \cap C)$ it must be in both $B \cap C$. Thus, $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Problem 1.6. *Show the following;*

$$B \cap \left(\bigcup_{i=1}^{\infty} X_i \right) = \bigcup_{i=1}^{\infty} B \cap X_i$$
$$\left(\bigcup_{i=1}^{\infty} X_i \right)^c = \bigcap_{i=1}^{\infty} X_i^c$$

Solution.

1. Let $x \in B \cap (\cup X_i)$. Since $x \in \cup X_i$ there exists at least one j such that $x \in X_j$. If $x \in B$ and $x \in X_j$ then there exists at least one non-zero entry such that $x \in B \cap X_j$. Thus, since x is general, we have that $B \cap (\cup X_i) = \cup (B \cap X_i)$.
2. Let $x \in \cup X_i$. There exists that for at least one integer j, $x \in X_j$. Thus, $x \notin X_j^c, \rightarrow x \notin \cap x_j^c$. Since x is general, we have that $(\cup X)^c = \cap X_i^c$.

1.2 Fields

In real analysis, one tends to discuss the real-number system. This system, although intuitive in early math courses, must be built up from an abstract sense.

Definition 1.4. *A field, F, is a set along with the operations of addition and multiplication. Let a, b, c be elements of F;*

1. *Commutativity:* $a + b = b + a$
2. *Additivity:* $(a + b) + c = a + (b + c)$
3. *Distributivity:* $a(b + c) = ab + ac$
4. *Additive Identity:* $a + 0 = a$
5. *Additive Inverse:* $a + (-a) = 0$
6. *Multiplicative Identity:* $1 \cdot a = a$
7. *Multiplicative Inverse:* $a \cdot a^{-1} = 1$

The rest is this section is treated as problems with proofs due to the nature of the subject.

Naturally, given one has proven that the additive inverse of 1 is -1 and that $-1 \cdot a = -a$, then one can define addition and subtraction;

1. Subtraction: $a - b$
2. Quotient: if $b \neq 0$, then $\frac{a}{b} = ab^{-1}$.

1.2.1 Ordered Fields

An ordered field is defined by the notion of $a < b$ where $a, b \in P$ where P defines a positive subset of a field.

Definition 1.5. *An ordered field, F, is a field with a positive subset P that abides by the following properties;*

1. *if $a \in F$ either $a = 0$, $a \in P$, or $-a \in P$.*
2. *if $a \in P$ then $-a \notin P$*
3. *if $a, b \in P$ then $a + b \in P$.*

In addition, ordered fields have several properties that arise from these. For example, that of multiplicative inverse.

Problem 1.7. *Show that if $a \in P$, $a^{-1} \in P$.*

Proof. Suppose that $a \in P$. Then, if $-a^{-1}$ were to be in P one can check by closure of multiplicative inverse. Specifically, we can see if $-1 = a(-a^{-1}) \in P$ is true. This is a contradiction by bullet 2 of Def. 1.5, which implies that a^{-1} must be in an ordered set. ■

The positive set P can then be used to define operations on less than or larger than:

Definition 1.6. *The less than or larger than symbols are defined under elements of P. Suppose F is an ordered field. Suppose $a, b \in F$;*

1. *if $a < b$ than $b - a \in P$*
2. *if $a \leq b$ than $a < b$ or $a = b$*
3. *if $a > b$ than $b < a$*
4. *if $a \geq b$ than $b < a$ or $a = b$*

We can then prove transitivity;

Problem 1.8. *Show that an ordered field F is transitive, that is, for elements $a, b, c \in F$, if $a < b$ and $b < c$ then $a < c$.*

Proof. By definition, $a < b$, then $b - a \in P$. Similarly, $c - b \in P$. Since P is closed under addition, we have that $b - a + (c - b) = c - a \in P$. By definition, $a < c$. ■

Next, we have the definition of absolute value.

Definition 1.7. *Suppose F is an ordered field and $b \in F$. The absolute value of b, denoted as $|b|$ is defined as:*

$$|b| = \begin{cases} b & \text{if } b \geq 0 \\ -b & \text{if } b < 0 \end{cases} \tag{1.9}$$

The absolute value is a useful tool that allows use to define and prove the existence of the triangle inequality;

Problem 1.9. *Suppose F is an ordered field and $a, b \in F$ show that*

$$|a + b| \leq |a| + |b| \,. \tag{1.10}$$

Proof. This can be shown in cases of the signs of each element.

1. $a, b > 0$; Then, $a + b \in P$ and by definition of the absolute value, $|a + b| = a + b = |a| + |b|$
2. $a, b < 0$; If both terms are less than 0, then we can use the distributive property of multiplication: $|a + b| = |-(a + b)| = a + b = |a| + |b|$.
3. $a < 0, b > 0$; Here, $a + b$ is not necessarily larger than 0. What matters is the output. If the sum of a, b is negative, it falls into (2.) and if positive it falls into (1.). Otherwise, one can write that if $b > a$, then $b + a < b$. By this logic, we have that $|b + a| < |b| < |b| + |a|$.
4. $b < 0, a > 0$; Same as (3.).
5. The logic of sum is if $a + b > 0$, then $|a + b| = a + b$. if $a + b < 0$, then $|a + b| = -(a + b) = (-a) + (-b) \leq |a| + |b|$.

■

Lastly, we can show that the ordered field of rational numbers $\mathbf{Q}$ as contained in each ordered field.

Definition 1.8. *Let $\varphi : \mathbf{Q} \to F$ such that $\varphi(0) = 0$ and;*

$$\varphi\left(\frac{m}{n}\right) = \frac{\sum_{i=1}^{m} 1}{\sum_{i=1}^{n} 1} \,. \tag{1.11}$$

As well as;

$$\varphi\left(-\frac{m}{n}\right) = \frac{\sum_{i=1}^{m} (-1)}{\sum_{i=1}^{n} 1} \tag{1.12}$$

Proof. First we show that this function is one-to-one, or surjective and bijective. Let m, n, p, q be positive integers. We have that $\varphi(m/n) = \varphi(p/q)$ if;

$$\begin{gathered} (\sum_{i=1}^{m} 1)(\sum_{i=1}^{q} 1) = (\sum_{i=1}^{p} 1)(\sum_{i=1}^{n} 1) \\ mq = pn \,. \end{gathered}$$

Thus, we have that $\frac{m}{n} = \frac{p}{q}$, which implies that for the rational functions this is fully one-to-one for all positive rational numbers.

For negative rational numbers, we can use a similar logic. Let m, p be negative rational numbers, we have that $-mq = -pn$ which shows that this function is one-to-one for all negative rational numbers. Thus, φ is one-to-one on all of $\mathbf{Q}$.

To verify that this map preserves properties of an ordered field, we can verify the properties one by one.

1. $\varphi(a+b) = \sum_{i=1}^{a+b} 1 = \sum_{i=1}^{a} 1 + \sum_{i=a+1}^{a+b} 1 = \sum_{i=1}^{a} 1 + \sum_{i=1}^{b} 1 = \varphi(a) + \varphi(b)$.
2. $\varphi(ab) = \sum_{i=1}^{ab} 1 = ab$. However, $\varphi(a) = a$ and $\varphi(b) = b$. so, $\varphi(a) \cdot \varphi(b) = ab = \varphi(ab)$.
3. $\varphi(-a) = \sum_{i=1}^{a} -1 = -1 \cdot a = -a = -1 \cdot \varphi(a)$.
4. $\varphi(a^{-1}) = \varphi(\frac{1}{a}) = \frac{1}{\sum_{i=1}^{a}} = \frac{1}{\varphi(a)}$.

■

This results means that we can treat $\mathbf{Q}$ as a subset of the ordered field F for all ordered fields.

1.2.2 Completeness

We start with the pythagorean theorem.

Problem 1.10. *Show that there is no rational number who's square is 2.*

Proof. Let m, n be integers such that there exists no common factor between them greater than 1. We aim to find a number $\left(\frac{m}{n}\right)^2 = 2$. Expanding the square, we have that $m^2 = 2n^2$. It is clear that m^2 is even. Thus, $m = 2k$ for some integer k. thus, $4k^2 = 2n^2$, or $2k^2 = n^2$. Thus n^2 is even as well. However, this is a contradiction as m and n are to have no common factors greater than 1. This is impossible if they are both even. Thus, there exists no m, n that satisfy the above. This shows that the square root of 2, $\sqrt{2}$ is a real number, that is not rational. ■

This shows that the ordered field of rational numbers, $\mathbf{Q}$, must have more properties to include a description of the real numbers.

Definition 1.9. *Suppose F is an ordered field and A is a subset of F; $A \subset F$. Then, let $b \in F$. The upper bound of A is defined as b such that for all $a \in A$, $a \leq b$.*

*The **least upper bound** is defined as $a \leq b$ such that;*

1. *b is an upper bound of A.*
2. *$b \leq c$ for all other upper bound $c \in F$.*

The existence of a least upper bound is so important that it has a name.

Definition 1.10. *An ordered field F is called complete if every non-empty subset of F that has an upper bound has a least upper bound.*

This is actually enough to define the real numbers.

Definition 1.11. *Let $\mathbf{R}$ define a complete ordered field. $\mathbf{R}$ is known as the field of real numbers.*

1.2.3 Problems

Problem 1.11. *Show that $a0 = 0$ for all a in a field F.*

Proof. Let $a \in F$. The element 0 is defined as the additive identity such that if

$$0 = 0\,, 0 + 0 = 0\,.$$

Using this property, one can use the distributivity of the field F to show that;

$$\begin{aligned} a0 &= a(0+0) \\ &= a0 + a0\,. \end{aligned}$$

Using the additive inverse,

$$0 = a0 \tag{1.13}$$

which concludes the proof. ■

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Problem 1.12. *Show the following. Let F be a field.*

- *For each $a \in F$, show that the additive inverse of a is unique.*
- *$-(-a) = a$ for each $a \in F$.*
- *$(-1)a = -a$ for each $a \in F$.*

Proof. The three proofs are shown in order.

- Let $a, b, c \in F$ such that $a + b = 0$ and $a + c = 0$. Then, if both of these equations are equal to 0, one can set them to be equal. That is,

  $$a + b = a + c\,.$$

  The additive inverse of a, which is $-a$ can be added to both sides, which implies that;

  $$b = c\,.$$

  However, since b is the additive inverse by definition, there is at most 1 additive inverse of a, which is defined as $-a$.

- Let $a \in F$. Let $b \in F$ such that $ab = (-a)$. Using the multiplicative inverse,

  $$\begin{aligned} ab \cdot a^{-1} &= (-a) \cdot a^{-1}\,, \\ &\text{distributivity} \\ baa^{-1} &= -(aa^{-1})\,, \\ b &= -1\,. \end{aligned}$$

  Thus, using the definition of the additive inverse, one finds that;

  $$\begin{aligned} a + (-a) &= 0\,, \\ -a + -(-a) &= -1 \cdot 0\,, \\ -a + -(-a) &= 0\,, \\ -a + a + -(-a) &= a\,, \\ -(-a) &= a\,. \end{aligned}$$

  which concludes the proof. We have also shown that since b is unique, there is a b that exists and $ba = -a$.

■

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1.3 The Real Number System

The real numbers are defined by a set of axioms. There are various ways to treat this. I am going to just state all of the properties upfront and then proceed with the consequences of their existence.

1.3.1 Axioms and Order

Theorem 1.3. ***Axioms of the Real Number System:*** *These are the axioms that define the set of real numbers. The properties are stated outright, then the consequences are listed in further statements.*

F1: ***Commutativity****; Let $a, b \in \mathbb{R}$, then $a + b = b + a$, and $a \cdot b = b \cdot a$.*

F2: ***Associativity****; Let Let $a, b, c \in \mathbb{R}$, then $(a + b) + c = a + (b + c)$, and $a \cdot (b \cdot c) = (a \cdot b) \cdot c$.*

F3: ***Distributitivity****; $a \cdot (b + c) = ab + ac$*

F4: ***Existence of Neutral Numbers****; $a + 0 = a$, $a \cdot 1 = a$*

F5: ***Existence of Additive and Multiplicative Inverse Numbers****; For any $a \in \mathbb{R}$ there exists a number b such that $a + b = 0$, where $b = -a$. Simiarily, there exists a number c such that $a \cdot c = 1$, where $c = a^{-1}$.*

O6: ***Order****; There is a subset of $\mathbb{R}$ known as $\mathbb{R}_+$ such that the two properties are satisified:*

- *If $a, b \in \mathbb{R}_+$, $a + b \in \mathbb{R}_+$, $a \cdot b \in \mathbb{R}_+$.*
- *For any $a \in \mathbb{R}$ only one of the following is true.*

$$a \in \mathbb{R}_+, \quad a = 0, \quad -a \in \mathbb{R}_+$$

.

The numbers such that $a \in \mathbb{R}_+$ are known as positive numbers, while the numbers such that $-a \in \mathbb{R}_+$ are known as negative numbers.

LUB7: ***Least Upper Bound****; A nonempty set of real numbers that is bounded from above will have a least upper bound.*

The real number system is completely defined by the Thm. 1.3. We can now form the properties and rules of algebra on the real number system.

1. In a sum or product of many numbers, the parentheses can be omitted; $(a + (b + c)) + d = a + ((b + c) + d) = (a + b) + (c + d) = \ldots$ by F1 and F2.
2. In a product of many numbers, order is immaterial; $a \cdot b \cdot c = b \cdot a \cdot c = c \cdot a \cdot b = \ldots$
3. For any $a, b \in \mathbb{R}$ the equation $x + a = b$ has one and only one solution. For, if $x \in \mathbb{R}$ then see the following computation.

$$\begin{aligned} x &= x + 0 \\ &= x + ((a) + (-a)) \\ &= (x + a) + (-a) \\ &= b + (-a) \\ x &= b + (-a) \end{aligned}$$

Thus, $x = b + (-a)$ is the only solution.

4. For any $a, b \in \mathbb{R}$, with exception $a \neq 0$ the equation $xa = b$ has one and only one solution.

$$\begin{aligned} x &= xaa^{-1} \\ &= (xa) \cdot a^{-1} \\ x &= ba^{-1} \end{aligned}$$

   Thus, the element 1 is unique from F4. we define $a^{-1} = \frac{1}{a}$

5. For any element $a \in R$, $a \cdot 0 + a \cdot 0 = a \cdot (0 + 0) = a \cdot 0 = a \cdot 0 + 0$. So both $a \cdot 0$ and 0 are solutions of the equation $x + a \cdot 0 = 0$. Thus, if a product of numbers is 0 then at least one of the element must be 0. for if $ab = 0$ and $a \neq 0$ then we can multiply both sides by a^{-1} to get $b = 0$. Hence, the illegitimacy of division by 0.

6. Solutions of the equation $x + (-a) = 0$ are given by $-(-a)$ given by (3.). By F5, we have $-(-a) + (-a) = 0$, so $-(-a) = a$ for any $a \in \mathbb{R}$.

7. $(a^{-1})^{-1} = a$ for any non-zero $a \in \mathbb{R}$. Since $a \cdot a^{-1} = 1$ by F5, and we know that $a^{-1} \neq 0$, then F4 implies that $(a^{-1})^{-1}$ and a are equal as both are solutions to $x + a^{-1} = 1$.

8. $-(a + b) = (-a) + (-b)$ for all $a, b \in \mathbb{R}$ as both are solutions to the equation $x + (a + b) = 0$.

9. $(ab)^{-1} = a^{-1}b^{-1}$ for any $a, b \in \mathbb{R}$, $a, b \neq 0$. Both are solutions to the equation $x(ab) = 1$.

   From this, we can define the rules of the fraction;

$$\begin{aligned} \frac{ac}{bc} &= ac(bc)^{-1} = ab^{-1}cc^{-1} = ab^{-1} = \frac{a}{b} \\ \frac{a}{b} \cdot \frac{c}{d} &= ab^{-1}cd^{-1} = ac(bd)^{-1} = \frac{ac}{bd} \\ \frac{a}{b} + \frac{c}{d} &= ab^{-1} + cd^{-1} = ab^{-1}dd^{-1} + cbb^{-1}d^{-1} \\ &= (ad + cb) \cdot (b^{-1}d^{-1}) = \frac{ad + cb}{bd} \end{aligned}$$

10. $a \cdot (-1) = -a$ for all $a \in \mathbb{R}$. Since $(-1) \cdot a + a = a((-1) + 1) = a \cdot 0 = 0$. We have that $(-1) \cdot a$ and a are solutions to the equation $x + a = 0$, hence are equal.

    A consequence of this is that $-a \cdot b = (-1) \cdot a \cdot b = -ab$ and $(-a) \cdot (-b) = (-1) \cdot (-1) \cdot a \cdot b = ab$

These are the field properties of the real numbers and apply to fields such as the rational and complex numbers. To define the reals, The order and LUB properties need to be applied. We can now discuss the order properties of the real numbers.

1. If $a, b \in \mathbb{R}$ only one of the following statements is true;

$$\begin{aligned} a &< b \\ a &= b \\ a &> b \end{aligned}$$

   This is the case as we apply $(a - b)$ then the number must fit in one of three subsets, being $a \in \mathbb{R}_+, -a \in \mathbb{R}_+, a \in \{0\}$.

2. if $a > b$ and $b > c$ then $a - b \in \mathbb{R}_+$ and $b - c \in R_+$. Thus, $a - c = (a - b) + (b - c) \in \mathbb{R}_+$. So, $a > c$.

3. If $a > b$ and $c \geq d$, then $a + c \geq b + d$. This implies that $a - b \in \mathbb{R}_+, c - d \in \mathbb{R}_+ \cup \{0\}$. Thus, $(a + c) - (b + d) = (a - b) + (c - d) \in \mathbb{R}_+$.

4. If $a > b > 0$ and $c \geq d > 0$, then $ac > bd$. For $a - b \in \mathbb{R}_+$ and $c \in \mathbb{R}_+$, then $ac - bc = (a - b)c \in \mathbb{R}_+$. Similarily, $c - d \in \mathbb{R}_+ \cup \{0\}$, $b \in \mathbb{R}_+$, and so $bc - bd \in \mathbb{R}_+ - \{0\}$. Thus, it follows that $ac - bd = (ac - bc) + (bc - bd) \in \mathbb{R}_+$, that is, $ac > bd$.

5. **Rules for addition and multiplication of real numbers;**
   - $(a_+) + (a_+) = (a_+)$
   - $(a_-) + (a_-) = (a_-)$
   - $(a_+) \cdot (a_+) = (a_+)$
   - $(a_-) \cdot (a_-) = (a_+)$
   - $(a_+) \cdot (a_-) = (a_-)$
6. For any $a \in \mathbb{R}, a^2 \in \mathbb{R}_+$. More tightly, $a^2 \geq 0$ where the equality is $0^2 = 0$ and special exceptions are $1^2 = 1$.
7. If $a > 0$, then $1/a > 0$.
8. If $a > b > 0$ then $1/b > 1/a$.

We can introduce exponentiation from the consequent multiplication of identical elements.

$$a^n = a \cdot a \cdot a \cdot \ldots$$

1.3.2 Absolute Value

The absolute value is defined as the following piecewise function;

$$|a| = \begin{cases} a & \text{if } a > 0 \\ 0 & \text{if } a = 0 \\ -a & \text{if } a < 0 \end{cases}$$

The absolute value has several properties;

1. $|a| \geq 0$ for all $a \in \mathbb{R}$ and $|a| = 0$ iff. $a = 0$.
2. $|ab| = |a| \cdot |b|$ for all $a \in \mathbb{R}$.
3. $|a|^2 = a^2$ for all $a \in \mathbb{R}$.
4. $|a + b| \leq |a| + |b|$ for all $a, b \in \mathbb{R}$.
5. $|a - b| \geq ||a| - |b||$ for all $a, b \in \mathbb{R}$.

The triangle inequality (4.) is of utmost importance as has been previously proved in Prf 1.2.1. See in Fig. 1.3.

1.3.3 Least Upper Bound Property.

The final concept for the real number system is the property of the least upper bound (LUB). If $S \subset \mathbb{R}$, then an upper bound for the set S is a number $a \in \mathbb{R}$ such that $s \leq a$ for each $s \in S$. If such a number exists, S is bounded from above. We call a real number y a **least upper bound** of the set S if y is an upper bound for S, and if a is any given upper bound for S, then $y \leq a$.

A non-empty, finite subset $S \subset \mathbb{R}$ will always have a LUB. We need to justify *LUB7*, which states that a non-empty set of real numbers, bounded from above, will always have a LUB.

The consequences of the LUB are listed;

1. For any real number x, there an an integer such that $n > x$.

   Proof. By contradiction, suppose there is a number that doesn't obey this. Then, $n \leq x$ and thus is bounded from above. Since The set of integers is non-empty and bounded from above by the number a. However, for each integer n, $n + 1$ is aso an integer. So, $n + 1 \leq a$. Thus, $n \leq a - 1$ is also an upper bound. Since $a - 1 < a$, $a - 1$ is not an upper boumd, hence this is a contradiction. ■

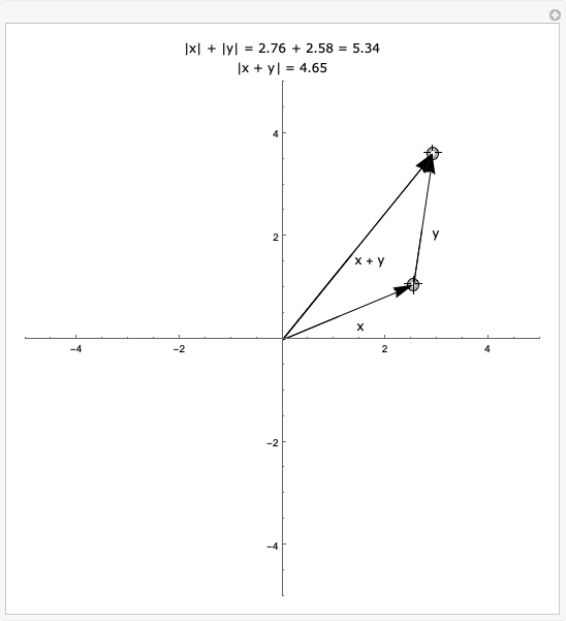


Figure 1.3: visualization of the triangle inequality; $|x+y| \leq |x| + |y|$ for all $x, y \in \mathbb{R}$.

2. For any positive number a there exists an integer n such that $\frac{1}{n} < a$.

   Proof. Since $a \in \mathbb{R}$, by the field property, $a^{-1} = 1/a \in \mathbb{R}$. Since we prove that for all real numbers, there exists $n > x$ we have that there exists integer $j > a$ such that $1/a > 1/j$ by the order property. Hence, $1/j > 0$. ■

3. For any $x \in \mathbb{R}$, there is an integer n such that $n \leq x < n + 1$.

   Proof. Let N be an integer such that $|N| > x$. Then, we have that x is bounded by the inequality $-N < x < N$. We have that this forms the set $\{-N, -N+1, \ldots, 0, 1, 2, \ldots, N\}$. We can choose n arbitraily along this such that $n \leq x < n + 1$ ■

4. For any $x \in \mathbb{R}$ and positive integer N there is an integer n such that

$$\frac{n}{N} < x < \frac{n+1}{N}$$

   Proof. We can first rewrite this equation by multiplying each set by N,

$$n < Nx < n + 1$$

   We need to show that $N \cdot x \in \mathbb{R}$. Since $x \in \mathbb{R}$ and $N \in \mathbb{Z} \subset \mathbb{R}$ we have that $x \cdot N \in \mathbb{R}$ by the field properties. Thus, let $y = Nx \in \mathbb{R}$, by (3.) we have

$$n \leq y < n + 1$$

   ■

5. if $x, a \in \mathbb{R}, a > 0$ then there exists a rational number r such that $|x - r| < a$. That is, we can approximate a real number as closely as we wish by a rational number.

   Proof. We first use (2.) to find a positive integer n such that $\frac{1}{n} < a$. By (4.) we can use that for x there exists $\frac{n}{N} < x < \frac{n+1}{N}$. Rewriting this by subtracting n/N;

$$0 < x - \frac{n}{N} < \frac{1}{N} < a$$

   Thus, We have that $|x - \frac{n}{N}| < a$ ■

1.3.4 Decimals

For a given set of integers, $a_1, a_2, \ldots a_n \in \{0, 1, 2, \ldots, 9\}$, we can write the decimal representation as

$$a_0.a_1a_2a_3\cdots = \frac{a_0}{10^0} + \frac{a_1}{10^1} + \frac{a_2}{10^2} + \cdots + \frac{a_n}{10^n}$$

Suppose $n, m \in \mathbb{Z}, m < n$. Then, we have the inequality that;

$$\begin{aligned} a_0.a_1a_2\ldots a_m &\leq a_0.a_1a_2\ldots a_n \\ &= a_0.a_1a_2\ldots a_m + a_{m+1}\cdot 10^{-(m+1)} + \cdots + a_n\cdot 10^{-n} \\ &\leq a_0.a_1a_2\ldots a_m + 9\cdot 10^{-(m+1)} + \cdots + 9\cdot 10^{-n} \end{aligned}$$

If we add a 1 to the last entry, it rounds up to the decimal in question;

$$a_0.a_1a_2\ldots a_m \leq a_0.a_1a_2\ldots a_n < a_0.a_1a_2\ldots a_m + 10^{-m}$$

Formally, the infinite decimal $a_0.a_1a_2\ldots$ is a representation of the least upper bound. Ie, every infinite decimal is a decimal expansion for a definite real number and we can use the infinite decimal iteself as a symbol for the number.

$$a_0.a_1a_2\cdots = \text{LUB}\{a_0.a_1a_2\ldots a_n : n \in \mathbb{Z}_+\} \tag{1.14}$$

and, for any positive integer n we have teh inequality

$$a_0.a_1a_2\ldots a_n \leq a_0.a_1a_2\cdots \leq a_0.a_1a_2\ldots a_n + 10^{-n}$$

Using this, we can write numbers that are not rational.

...

1.3.5 Square Roots

We will discuss the Schwarz inequality as it is useful later;

Definition 1.12. *The Schwarz inequality is defined as*

$$|a_1b_1 + a_2b_2 + \cdots + a_nb_n| \leq \sqrt{a_1^2 + a_2^2 + \cdots + a_n^2}\sqrt{b_1^2 + b_2^2 + \cdots + b_n^2} \tag{1.15}$$

Proof. We start with the fact that

$$0 \leq (\alpha a_1 - \beta b_1)^2 + (\alpha a_2 - \beta b_2)^2 + \cdots + (\alpha a_n - \beta b_n)^2 \tag{1.16}$$

the goal of writing this equation is to factor it out into the RHS. We can multiple each term out;

$$\begin{aligned} 0 &\leq (\alpha a_1 - \beta b_1)^2 + (\alpha a_2 - \beta b_2)^2 + \cdots + (\alpha a_n - \beta b_n)^2 \\ &= \alpha^2(a_1^2 + a_2^2 + \ldots) + \beta^2(b_1^2 + b_2^2 + \ldots) - 2\alpha\beta(a_1b_1 + a_2b_2 + \ldots) \end{aligned}$$

We can let $\beta = \sqrt{a_1^2 + a_2^2 + \ldots}, \alpha = \sqrt{b_1^2 + b_2^2 + \ldots}$ to add the square terms together;

$$\begin{aligned} 0 &\leq 2(b_1^2 + b_2^2 + \ldots)(a_1^2 + a_2^2 + \ldots) - 2\alpha\beta(a_1b_1 + a_2b_2 + \ldots) \\ (a_1b_1 + a_2b_2 + \ldots) &\leq \frac{1}{\alpha\beta}(b_1^2 + b_2^2 + \ldots)(a_1^2 + a_2^2 + \ldots) \\ &= \sqrt{a_1^2 + a_2^2 + \ldots}\sqrt{b_1^2 + b_2^2 + \ldots} \end{aligned}$$

or,

$$|a_1b_1 + a_2b_2 + \cdots + a_nb_n| \leq \sqrt{a_1^2 + a_2^2 + \cdots + a_n}\sqrt{b_1^2 + b_2^2 + \cdots + b_n} \tag{1.17}$$

■

As a corollary, note that using the property of the absolute value,

Definition 1.13.

$$\sqrt{\sum_j (a_j - b_j)^2} \leq \sqrt{\sum_j a_j^2} + \sqrt{\sum_j b_j^2} \tag{1.18}$$

Proof.

$$\begin{aligned}
\sum_j (a_j - b_j)^2 &= \sum_j a_j^2 + b_j^2 - 2a_j b_j \\
&= (a_1^2 + \dots) + (b_1^2 + \dots) - 2(a_1 b_1 + a_2 b_2 + \dots) \\
\text{Schwarz} \quad &\leq (a_1^2 + \dots) + (b_1^2 + \dots) + 2\sqrt{a_1^2 + \dots}\sqrt{b_1^2 + \dots} \\
&= \left[\sqrt{a_1^2 + \dots} + \sqrt{b_1^2 + \dots}\right]^2
\end{aligned}$$

Thus, we have shown that;

$$\sqrt{\sum_j (a_j - b_j)^2} \leq \sqrt{a_1^2 + a_2^2 + \dots + a_n^2} + \sqrt{b_1^2 + b_2^2 + \dots + b_n^2} \tag{1.19}$$

■

1.3.6 Problems

1.4 Metric Spaces

A metric space is defined as a set E with a rule $d(q,p)$ $\forall q, p \in E$. If d satisfies

$$\begin{aligned}
&d(q,p) \geq 0, \\
&d(q,p) = 0, \quad \text{iff. } q = p, \\
&d(q,p) = d(p,q), \\
&d(q,r) \leq d(q,p) + d(p,r).
\end{aligned} \tag{1.20}$$

Example 1.2. *Let $E = \mathbb{R}$ with rule $d(q,p) = |q - p|$. The first three properties are trivial.*

Proof. *We aim to show that*

$$|q - r| \leq |q - p| + |p - r|$$

This proof utilizes the property of the absolute value;

$$|a + b| \leq |a| + |b| \tag{1.21}$$

Thus,

$$\begin{aligned}
|q - r| &= |q + p - p - r| \\
&= |(q - p) + (p - r)| \\
&\leq |(q - p)| + |(p - r)| \\
&= |q - p| + |p - r|
\end{aligned}$$

■

Thus, $E = \mathbb{R}$ is a metric space.

Example 1.3. *Let $E^n = \mathbb{R}^n$ with rule $d(x,z) = \sqrt{(x_1 - z_1)^2 + (x_2 - z_2)^2 + \dots + (x_n - z_n)^2}$.*

Proof. *As before the first three properties are trivial. The triangle inequality is proven by using the Schwarz inequality and it's corollary;*

$$\begin{aligned}\sqrt{\sum_j (x_j - z_j)^2} &= \sqrt{\sum_j (x_j - p_j + p_j - z_j)^2} \\ &= \sqrt{\sum_j ((x_j - p_j) + (p_j - z_j))^2} \\ &= \sqrt{\sum_j (a_j + b_j)^2} \\ &\leq \sqrt{\sum_j a_j^2} + \sqrt{\sum_j b_j^2} \\ &= \sqrt{\sum_j (x_j - p_j)^2} + \sqrt{\sum_j (p_j - z_j)^2}\end{aligned}$$

This shows;

$$d(x, z) \leq d(x, p) + d(p, z) \tag{1.22}$$

■

Thus, E^n is a metric space.

1.5 Polynomials

The polynomial function is defined as a map consisting of a sum of indeterminant variables x, meaning they are not yet known or determined, raised to an positive integer power. The degree of the polynomial is defined by the power leading of the leading term. Let $x \in \mathbb{R}$ and f be a map. The general polynomial function of the n-th degree is defined as;

$$f(x) = \sum_{j=0}^{n} a_j x^j \tag{1.23}$$

where a_j are real-valued constants. The degree of n fixes the leading constant to be non-zero, $a_n \neq 0$.

The polynomials of first and second degree, known as the linear and quadratic forms, are most famous and widely used. These functions are used so much that they are taught in high-school as introductory mathematics.

Definition 1.14. *Let $x \in \mathbb{R}$ and the constants a_1 and $a_0 \in \mathbb{R} \setminus 0$. The linear map, $f : \mathbb{R} \to \mathbb{R}$ is defined as*

$$f(x) = a_1 x + a_0 \tag{1.24}$$

where a_1 is defined as the slope of the line, and a_0 is defined as the intercept of the line.

Definition 1.15. *Let $x \in \mathbb{R}$ and the constants a_2, a_1, and $a_0 \in \mathbb{R}$. The standard-form of the quadratic map, $f : \mathbb{R} \to \mathbb{R}$ is defined as*

$$f(x) = a_2 x^2 + a_1 x + a_0 \tag{1.25}$$

where $a_2 \in \mathbb{R} \setminus 0$ and a_1 and $a_0 \in \mathbb{R}$. The quadratic form can be factored in various ways to describe different properties of it. The vertex-form is defined as

$$f(x) = a_2 (x - h)^2 + k \tag{1.26}$$

where the substitutions h and k define the position of the vertex. These constants, easily derived using a derivative, are given as

$$h = -\frac{a_1}{2a_2}\,, \quad k = a_0 - \frac{a_1^2}{4a_2}\,. \tag{1.27}$$

The solutions of a univariate polynomial, known as the roots, exist if and only if $f(z) = 0$ for a polynomial solution z. If a polynomial has N solutions z_i, the form of the factored polynomial simply exists as

$$f(x) = c(x - z_n)(x - z_{n-1}) \dots (x - z_2)(x - z_1) \tag{1.28}$$

where z_i can be a complex number. In the case of positive coefficients only, there exists at most n solutions.

To factor a polynomial,

$$f(x) = a_n x^n a_{n-1} x^{n-1} + \cdots + a_1 x + a_0\,, \tag{1.29}$$

the rational root test can be used where $a_i \in \mathbf{Z}$ and $a_n, a_0 \neq 0$. The rational root theorem states that candidate numbers q can be defined as

$$q = \frac{p}{q}\,, \tag{1.30}$$

such that p is an integer factor of a_0 and q is an integer factor of a_n. Thus, all possible roots consist candidates q for all factors p, q.

The applications of the polynomials are wide. For example, see Gaussian integrals, Legendre polynomials, and free-energy landscapes, where quadratic solutions are generally made to ensure Gaussianeity.

1.5.1 Binomial Theorem

The binomial theorem is a way to expand binomials;

$$(x + y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k\,. \tag{1.31}$$

It is useful.

Chapter 2

Measure Theory

Measure theory, in a broad sense, is the theory of how to define measurements, or actions upon the length of something.

2.1 Measurable Spaces

A measurable space is a pair of a set E and a set of subsets of E which we will refer to as ε. We first consider the set E as a mathematical object. We would like do perform mathematical operations on this. The set, given that it allows these operations, should be closed under them. This is the fundamental motivation for a measurable space.

Consider then the set E being finite. We would like to ensure that this set is closed under finite union. This implies that for a subset $A_i \subset E$,

$$\forall A_i \in E, \quad \bigcup_i A_i \in E\,. \tag{2.1}$$

This implies that the unions of two subsets will form a new subset which is already contained in E. This idea is defined in detail as an *algebra*.

Definition 2.1. *An algebra on the set E is defined as a non-empty collection of sets ε which is closed under finite union and complement.*

$$\begin{aligned} \text{if} A \in \varepsilon, \quad \text{then} \quad A^c \in \varepsilon\,, \\ \text{if} A, B \in \varepsilon, \quad \text{then} \quad A \cap B \in \varepsilon\,. \end{aligned}$$

For a finite set E, the set itself and its complement must be included in the algebra;

$$E \in \varepsilon, \quad \emptyset \in \varepsilon\,. \tag{2.2}$$

2.1.1 Sigma-algebras

A sigma-algebra is a special kind of algebra which is closed under complements and countable unions.

Definition 2.2. *A non-empty collection of subsets σ_E, which consist of measurable sets E_i, is known as a sigma-algebra on E if*

1. $A \in \sigma_E, \quad \to E \setminus A \in \sigma_E\,,$
2. $A_1, A_2, \cdots \in \sigma_E, \quad \to \bigcup_n A_n \in \sigma_E\,.$

Some properties are that every sigma-algebra on E contains at least E and $\emptyset$. That is the simplest sigma-algebra is $\sigma_E = \{E, \emptyset\}$.

Since a sigma-algebra is fundamentally a collection of sets, one can intersect a family of sigma-algebras on E, which when generates again a sigma-algebra on E.

For example, consider an arbitrary collection of subsets of E, $\mathcal{C}$. Then, a sigma-algebra which contains $\mathcal{C}$ can be generated. Suppose that $\mathcal{C}$ contains finite countable elements C_i, then the sigma algebra would be constructed by the following;

$$\sigma = \{E, \emptyset, C_i, C_i \cup C_j, \dots, C_i^c, \dots\}, \tag{2.3}$$

which specifically consists of all elements of $\mathcal{C}$, as well as ensuring closure under complement and union.

Another such example is known as the power set, or the sigma-algebra consisting of each and every possible subset of E. To imagine this, consider a partition into 3 components. The explicit power set is then;

$$2^E = \{C_1 \cup (C_2 \cup C_3), C_1 \cup C_2, C_1 \cup C_3, C_2 \cup C_3, C_1, C_2, C_3, \emptyset\}. \tag{2.4}$$

The set E contains 3 partitions, which is expressed as $2^3 = 8$ elements in the power set. The proof for this can be done via induction. An intuitive explanation is the fact that the set operation $\cup$ acts between 2 elements without parity. Thus, for all possible unions of 3 elements, there are 3 possible combinations. For 4 elements, there are 6 combinations. in addition, one must add the higher order additions as well. As one can see, this is related to the sum of all subsets with k connections.

Given a set of n elements, the size of the power set will consist of the sum of all possible subsets. This contains the combinatoric collection of unions between each of union-orders. By the binomial theorem;

$$\sum_{k=0}^{n} \frac{n!}{k!(n-k)!} = (1+1)^n = 2^n. \tag{2.5}$$

Returning to $\mathcal{C}$, if one takes all sigma-algebras that contain $\mathcal{C}$ and intersect them, they will arrive at the smallest possible sigma-algebra that contains $\mathcal{C}$, which is known as the generator of $\mathcal{C}$ and is denoted $\sigma\mathcal{C}$.

Remarks; the countable intersection of two (or more) sigma-algebra is again a sigma-algebra. The the union of two (or more) sigma-algebras is not necessarily a sigma-algebra.

Proof. By example, let Ω be a set and $A, B \subset \Omega$. The sigma-algebra $\mathcal{F}_i$ consists of

$$\mathcal{F}_1 = \{\emptyset, \Omega, A, A^c\}, \tag{2.6}$$

and

$$\mathcal{F}_2 = \{\emptyset, \Omega, B, B^c\}. \tag{2.7}$$

The intersection of the two is simply the $\{\emptyset, \Omega\}$ is the simplest sigma-algebra by definition.

The union of the two, $\mathcal{F}_3 = \mathcal{F}_1 \cup \mathcal{F}_2$;

$$\mathcal{F}_3 = \{\emptyset, \Omega, A, A^c, B, B^c\}, \tag{2.8}$$

is not closed under finite union or complement as $A \cup B \notin \mathcal{F}_3$. ■

It should be noted that the **uncountable intersection** of sigma-algebras are also sigma-algebras.

2.1.2 p, d systems

A collection of subsets $\mathcal{P}$ is a p-system if it is closed under intersection.

A collection of subsets $\mathcal{D}$ is a d-system if it satisfies the following properties;

1. $E \in \mathcal{D}$,
2. $A, B \in \mathcal{D}$ and $A \supset B$ then $A \setminus B \in \mathcal{D}$,
3. $(A_n) \subset \mathcal{D}$ and $A_n \nearrow A$ then $A \in \mathcal{D}$,

The notation (A_n) denotes a sequence. The notation $A_n \nearrow A$ implies that as the sequence continues, the limit of the sequence approaches the value of A.

Definition 2.3. *A collection of subsets of E is a sigma-algebra if and only if it is both a p-system and a d-system on E.*

Proof. *To prove this statement, it is trivial that a sigma-algebra satisfies all of the criteria of both p and d systems. Thus, it is our task to show the converse.*

Let the collection $\mathcal{E}$ on E be both a p-system and a d-system. It is closed under intersection. We require for it to be a sigma-algebra that is it closed under finite union and complement.

Since $\mathcal{E}$ is a d-system, it contains E and is closed under set subtraction. By clause 2 of the d-system,

$$A \in \mathcal{E}, \rightarrow E \setminus A \in \mathcal{E}\,,$$

which implies that $E^c \in \mathcal{E}$. This can be repeated for any element $E_i = E_k \cup E_j, E_k, E_j \in \mathcal{E}$.

Since $\mathcal{E}$ is closed under complement and intersection $(A^c \cap B^c) \in \mathcal{E}$. Thus, since this is closed under complement, then the complement is in $\mathcal{E}$:

$$(A^c \cap B^c) = (A \cup B)\,, \tag{2.9}$$

which is precisely closure under union by the property of being a p-system.

This idea is extended to finite unions as given a sequence $(A_n) \subset \mathcal{E}$, then one can define a sequence $B_1 = A_1, B_2 = A_1 \cup A_2, \ldots$. Since here $B_n \nearrow \bigcup_n A_n$, then by clause 3 of the d-system, $\bigcup_n A_n \in \mathcal{E}$. ■

. . . . . .

Definition 2.4. *Let $\mathcal{D}$ be a d-system on E. Fix D in $\mathcal{D}$ and let*

$$\hat{\mathcal{D}} = \{A \in \mathcal{D} : A \cap D \in \mathcal{D}\}\,. \tag{2.10}$$

Then, $\hat{\mathcal{D}}$ is again a d-system.

Proof. To show that $\hat{\mathcal{D}}$ is a d-system on E it is sufficient to check all of its requirements. Firstly, the set notation is written as all elements A in $\mathcal{D}$ such that $A \cap D \in \mathcal{D}$. Naturally, this implies that $E \in \hat{\mathcal{D}}$. $E \in \mathcal{D}, E \cap D = D, D \in \mathcal{D}$.

To show set subtraction on $\hat{\mathcal{D}}$, note that for all $A, B \in \mathcal{D}$ such that $A \supset B$, $A \setminus B \in \mathcal{D}$. By definition, all elements with an intersection between A and D are in the set $\hat{\mathcal{D}}$. This implies that all elements $A, B \in \hat{\mathcal{D}} \supset \mathcal{D}$. Since $A, B \in \mathcal{D}$, then $A \setminus B \in \mathcal{D}$ by the definition of a d-system.

Lastly, consider a sequence (A_n). It is sufficient to show that the union of this sequence is contained in $\hat{\mathcal{D}}$.

$$\begin{aligned}\hat{\mathcal{D}} &= \{\cup_n A_n \in \mathcal{D}\} \& \{\cup_n A_n \cap D \in \mathcal{D}\} \\ &= \{\cup_n A_n \in \mathcal{D}\} \& \{\cup_n (A_n \cap D) \in \mathcal{D}\}\end{aligned}$$

Since $A_n \cap D \in \mathcal{D}$ by the property of the d-system, then $A \in \hat{\mathcal{D}}$. ■

The next property is known as the Monotone class theorem.

Definition 2.5. *If a d-system contains a p-system, it also then contains the sigma-algebra generated by that p-system.*

Proof. *Let $\mathcal{C}$ be a p-system. Let $\mathcal{D}$ be the smallest d-system on E that contains $\mathcal{C}$. We want to prove $\mathcal{D} \supset \sigma\mathcal{C}$. It is sufficient to show that $\mathcal{D}$ is a sigma-algebra, however, since a d-system is a sigma-algebra if and only if it is both a d and p system, it is necessary to show that $\mathcal{D}$ is also a p-system.*

We aim to show closure under intersection.

Let B be fixed in $\mathcal{C}$;

$$\mathcal{D}_1 = \{A \in \mathcal{D} : A \cap B \in \mathcal{D}\}.$$

By the previous lemma, since $B \in \mathcal{D}$, $\mathcal{D}_1$ is a d-system. Similarly, for a fixed $A \in \mathcal{D}$,

$$\mathcal{D}_2 = \{B \in \mathcal{D} : A \cap B \in \mathcal{D}\},$$

which is a d-system as $B \in \mathcal{D}$. The key here is that since $\mathcal{C}$ is contained in $\mathcal{D}$, the set B is in $\mathcal{D}$. Also, $\mathcal{D}_1$ is a d-system that contains $\mathcal{C}$ if $A \in \mathcal{C}$ and $B \in \mathcal{C}$ then $A \cap B \in \mathcal{C}$ under closure. Thus, $\mathcal{D}_1$ must be the smallest d-system containing $\mathcal{C}$, or $\mathcal{D}_1 \supset \mathcal{D}$. In other words, $A \cap B \in \mathcal{D}$ for all $A, B \in \mathcal{C}$.

Thus, $A \cap B \in \mathcal{D}$ whenever A and B are in $\mathcal{D}$. This implies closure under intersection, which implies $\mathcal{D}$ is a p-system. ■

2.1.3 Problems

Problem 2.1. *Prove that if ε is an algebra on E, then*

1. $E \in \varepsilon$
2. $\emptyset \in \varepsilon$

Solution. Since ε is a non-empty set, and is an algebra, it is closed under finite union. Thus, since E is a finite set, one can partition E into a finite number of elements E_i. Under the closure of finite union,

$$\bigcup_i E_i \subseteq E, \tag{2.11}$$

which is necessarily an element of ε.

Another proof here is that given E is non-empty, E_i exists. By the definite of ε the complement of this is also in the set ε. Thus, under the finite closure of the union, $E_i \cup E_i^c \in \varepsilon$. Note here that $E_i \cup E_i^c = E$, which implies that $E \in \varepsilon$.

Similarly, single the algebra is closed under finite complement, one can use the fact that $E \in \varepsilon$;

$$\begin{aligned} E &\in \varepsilon \\ E^c &\in \varepsilon \\ E^c &= \emptyset \\ \emptyset &\in \varepsilon \end{aligned}$$

Problem 2.2. *Partition generated by a σ-algebras.*

1. *Let $\mathcal{C} = \{A, B, C\}$ be a partition of E List all the elements of $\sigma\mathcal{C}$*
2. *let $\mathcal{C}$ be a countable partition of E. show that every element of $\sigma\mathcal{C}$ is a countable union of elements taken from $\mathcal{C}$.*
3. *Let $E = \mathbb{R}$. Let $\mathcal{C}$ be the collection of all singleton subsets of $\mathbb{R}$, that is, sets that only contain a single point. Show that each element of $\sigma\mathcal{C}$ is either countable, or the complement of a countable set.*

Solution.

1. The elements of $\sigma\mathcal{C}$ are defined as the intersection of all sigma-algebras that contain $\mathcal{C}$. Since $\mathcal{C}$ contains 3 subsets of E, as it is a partition, which implies that A, B, C are disjoint and have a union of E. Thus, we can define $\sigma\mathcal{C}$ as the following set:

$$\sigma = \{\emptyset, E, \{A\}, \{B\}, \{C\}, \{A \cap B\}, \{A \cap C\}, \{B \cap C\}\} \tag{2.12}$$

This set contains E, all partitions of closures by E generated from $\mathcal{C}$. Thus, this is the smallest sigma-algebra that contains E generated by $\mathcal{C}$.

2. Since $\mathcal{C}$ is a partition, $\bigcup \mathcal{C} = E$. We need to show that $\sigma\mathcal{C}$ contains only elements from C. We can take ε as the set of all countable unions taken from E. We can show that this is a sigma-algebra by taking its definitions. Since ε contains E the subset of all the partitions, it must also contain the union of none of the partitions, or $\emptyset$. Secondly, by taking all of the unions, we start with the disjoint partitions and then create their unions until we reach E. This will be closed under complement as by taking countable unions, we will have the sets that are not part of the union as another subset of a different union. Thus, we can argue that this is the same as $\sigma\mathcal{C}$ by the logic of construction. Thus, since each element of $\sigma\mathcal{C}$ is isomorphic to the construction that we obtained from the union of subsets, we have shown that all elements are just a set of unions in $\mathcal{C}$.

3. Define the set $\mathcal{A} = \{A \subset \mathbb{R} : A \text{ is countable or } A^c \text{ is countable}\}$. We want to show that this is a sigma-algebra. Since $\emptyset$ is countable, it is contained within $\mathcal{A}$. $\emptyset \in \mathcal{A}$. The complement of this is the set of reals; $\mathbb{R} = \emptyset^c \in \mathcal{A}$. By definition, if A is countable, then A^c is co-countable. Similarly, if A is co-countable, this implies that A^c is countable. Thus, both sets are contained in $\mathcal{A}$.

Thus, consider a sequence $(A_n)_{n\in\mathbb{N}} \subset \mathcal{A}$. the union of this sequence is $\bigcup_n A_n$. This is split into cases. First, if all A_n are countable, so it the union, which is included in A_n. If all A_n are co-countable, then consider the compliment of the union $(\bigcup_n A_n)^c = \bigcap_n A_n^c$ which is an intersection of countable sets, which is countable. If there is a mix between countable and co-countable sets, they are co-countable, which are contained in A by the previous definition. Thus, $\bigcup_n A_n \in \mathcal{A}$. Thus, $\mathcal{A}$ is closed under union and complement. Thus, $\mathcal{A}$ is a sigma-algebra. Thus, $\sigma\mathcal{C} \subset \mathcal{A}$

We want to show that any countable set is a union of the singletons. Since A contains countable and co-countable sets, A is a union of singletons, which are by definition contained in $\mathcal{A}$.

$$A = \bigcup_{x\in A} \{x\} \tag{2.13}$$

For co-countable sets in A, we note that the complements are countable, which are contained in the above definition. Since $\mathcal{A}$ consists of countable sets, these are the countable unions of singletons contained in $\sigma\mathcal{C}$. Since $\mathcal{A}$ also consists of co-countable sets, these are the complements of intersections of countable sets in $\sigma\mathcal{C}$. Thus, $\mathcal{A} \subset \sigma\mathcal{C}$.

. . . . . .

Problem 2.3. *Let (E, ε) be a measurable space. Fix $D \subset E$ and let*

$$\mathcal{D} = \varepsilon \cap D = A \cap D : A \in \varepsilon$$

Show that $\mathcal{D}$ is a sigma-algebra on D. It is called the trace of (E, ε) on D.

Solution. To show that $\mathcal{D}$ is a sigma-algebra, we need to show closure under union, complement, intersection, and the existence of $D, \emptyset \in \mathcal{D}$.

1. To show that $D, \emptyset \in \mathcal{D}$ i will write out the logical steps. Since $D \subset E$, $D \in \epsilon$. Since $A = D, D \cap D = D$. By defintion of $\mathcal{D}$, $D \in \mathcal{D}$. Similarily, $\emptyset \subset E$, $\emptyset \in \epsilon$. Since $A = \emptyset, \emptyset \cap D = \emptyset$. By defintion of $\mathcal{D}$, $\emptyset \in \mathcal{D}$.

2. The closure of complement can be shown by the set property of D^c. Since $E^c = \emptyset$, $D^c = E\ D$.

$$\begin{aligned}(A_n \cap D)^c &= A^c \cup D^c \\ &= A^c \cup (E\ D) \\ &= E\ (D\ A) \\ &\in \mathcal{D}\end{aligned}$$

Thus $(A_n \cap D)^c \in \mathcal{D}$.

3. For union and intersection we will use a similar set operation;

$$\begin{aligned}\left(\bigcup_{i=1}^{\infty} A_i\right) \cap D &= \lim\left(A_1 \cup A_2 \cup \cdots \cup A_n\right) \cap D \\ &= \lim(A_1 \cap D) \cup (A_2 \cap D) \cup \cdots \cup (A_n \cap D) \\ &= \bigcup_{i=1}^{\infty}(A_i \cap D) \\ &\in \mathcal{D}\end{aligned}$$

$$\begin{aligned}\left(\bigcap_{i=1}^{\infty} A_i\right) \cap D &= (A_1 \cap A_2 \cap \cdots \cap A_n) \cap D \\ &= \lim(A_1 \cap D) \cap (A_2 \cap D) \cap \cdots \cap (A_n \cap D) \\ &= \bigcap_{i=1}^{n}(A_i \cap D) \\ &\to \left(\bigcap_{i=1}^{\infty}(A_i \cap D)\right)^c \\ &= \bigcup_{i=1}^{\infty}(A_i \cap D)^c \\ &= \bigcup_{i=1}^{\infty}(A_i^c \cup D^c)\end{aligned}$$

By the closure of the complement: $\in \mathcal{D}$

Thus we have shown that $\mathcal{D}$ is a sigma-algebra on D.

. . . . . .

Problem 2.4. *Let (E, ε) be a measurable space, let $\{\Delta\}$ be a single-point not in E. Let $\bar{E} = E \cap \{\Delta\}$. Show that*

$$\hat{\varepsilon} = \varepsilon \cup \{A \cup \{\Delta\} : A \in \epsilon\}$$

. . . . . .

Problem 2.5. *Let (E, ε), $(F, \mathcal{F})$ be measurable spaces. Show that the product $\varepsilon \otimes \mathcal{F}$ is also the sigma-algebra generated by $\hat{\varepsilon} \cup \mathcal{F}$ where*

$$\hat{\varepsilon} = \{A \times F : A \in \epsilon\} \qquad \mathcal{F} = \{E \times A : A \in \mathcal{F}\} \tag{2.14}$$

2.2 Measurable Functions

Functions from a set E to F is a rule that assigns an element $f(x)$ of F to each x in E; $f : E \to F$. Given a subset B of F, the inverse image of B under f:

$$f^{-1}B = \{x \in E : f(x) \in B\}. \tag{2.15}$$

The properties of these functions are as follows;

Definition 2.6. *Properties of the set function;*

1. $f^{-1}\emptyset = \emptyset$.
2. $f^{-1}F = E$.
3. $f^{-1}B \setminus C = (f^{-1}B) \setminus (f^{-1}C)$.
4. $f^{-1}\bigcup_i B_i = \bigcup_i f^{-1}B_i$
5. $f^{-1}\bigcap_i B_i = \bigcap_i f^{-1}B_i$

Proof.

1. *Since $\emptyset$ has no elements, there is no $f(x)$ in $\emptyset$. Thus, there is no such x that satisfies this.*
2. *Since f is a map from $E \to F$, each value of F has an inverse mapping to some element in E. This mapping need not be bijective, however, it is surjective. Thus, The inverse set of F must be E by construction.*
3. *Let $x \in f^{-1}B \setminus C$. Then, $f(x) = B \setminus C$. Here $f(x) \in B$, and $f(x) \notin C$. So define the inverse image, $f^{-1}B = \{x \in E : f(x) \in B\}$, and $f^{-1}C = \{x \in E : f(x) \in C\}$. It is apparent that $x \in f^{-1}B$. Similarly, $x \notin f^{-1}C$ as $B \setminus C \notin C$. Thus, $x \in f^{-1}B$ and $x \notin f^{-1}B$. So $f^{-1}B \setminus C = f^{-1}B \setminus f^{-1}C$.*
4. *For all $f(x) = \cup_i B_i$ there exists a pre-image $y = f^{-1}(\cup_i B_i)$. Thus, there exists an i such that $y \in B_i$. This implies that if there is at least 1 i that conditions this, then there is at least 1 pre-image $y = f^{-1}B_i$. Then, to ensure all values on E are contained, $y = \bigcup_i f^{-1}B_i$. Hence, $f^{-1}\bigcup_i B_i = \bigcup_i f^{-1}B_i$.*
5. *Since for all $f(x) = \cap_i B_i$ then $f(x) \in B_i$ for all i. This implies that the pre-image, y such that $y = f^{-1}(\cap_i B_i)$ is contained within all each and every $f(x)$ simultaneously. That is $y = \cap_i f^{-1}(B_i)$. Thus $f^{-1}(\cap_i B_i) = \cap_i f^{-1}(B_i)$.*

■

Measurable functions are defined as maps between measurable spaces. Let $(E, \mathcal{E})$ and $(F, \mathcal{F})$ be measurable spaces. A measurable function is defined as

$$f : E \to F, \quad f^{-1}B \in \mathcal{E} \quad \forall B \in \mathcal{F} \tag{2.16}$$

In english, this is stating that for some function f from E to F there exists a pre-image f^{-1} that maps from some element B in $\mathcal{F}$ to some element in $\mathcal{E}$. Or that there is a pre-image map between the subsets in the sigma-algebras of each of the spaces.

In order for $f : E \to F$ to be measurable, it is necessary that for some collection $\mathcal{F}_0$ that generates $\mathcal{F}$, meaning that the intersection of all sigma-algebras containing $\mathcal{F}_0$ is equal to $\mathcal{F}$, we have that $f^{-1}B \in \mathcal{E}$ for all $B \in \mathcal{F}_0$.

In english, if one wants to test that a function is measurable, it is only necessary to test on a subset that generates it.

Proof. Why is this necessary? Suppose that there is no such $\mathcal{F}_0$. If the sigma-algebra then does not exist then $\mathcal{F}$ is not the smallest sigma-algebra that exists on E. This fact implies that F is not a measurable space.

Sufficiency of this claim is now proved. Let $\mathcal{F}$ be the smallest sigma-algebra generated by $\mathcal{F}_0$. Then, we need to show that the inverse of E is contained within $\mathcal{F}$. Here,

$$\mathcal{F}_1 = \{B \in \mathcal{F} : f^{-1}B \in \mathcal{E}\}\,. \tag{2.17}$$

Thus, we need to show that $\mathcal{F}_1$ is a sigma-algebra. By the properties of functions, this map is obviously closed under union and since it contains the sample space, it is also closed under complement via the property of set subtraction. ■

The composition of functions is defined by

$$g \circ f(x) = g(f(x)), \quad x \in E\,, \tag{2.18}$$

for a mapping $f : E \to F$ and a mapping $g : F \to G$. The inverse composition is defined by

$$(g \circ f)^{-1}(x) = f^{-1}(g^{-1}(x)), \quad x \in G\,. \tag{2.19}$$

We can prove that measurable functions of measurable functions are measurable. Let f be a measurable function on E from $\mathcal{E} \to \mathcal{F}$. Let g be a measurable function on F from $\mathcal{F} \to \mathcal{G}$. Then, $(g \circ f)$ is measurable from $\mathcal{E} \to \mathcal{G}$;

Proof. The inverse, $(g \circ f)^{-1}$, can be expanded and acted on some $C \in \mathcal{G}$;

$$\begin{aligned}(g \circ f)^{-1}C &= f^{-1}(g^{-1}C)\,,\\ &= f^{-1}(B), \quad \text{where} B \in \mathcal{F}\,,\\ &= A, \qquad \text{where} A \in \mathcal{E}\,.\end{aligned}$$

Thus, $g \circ f$ is measurable. ■

2.2.1 Numerical Functions

A numerical function in measure theory is a function whose codomain is a set of numbers (such as real numbers $\mathbb{R}$, extended real numbers $\overline{\mathbb{R}}$, or complex numbers $\mathbb{C}$), rather than general abstract spaces. Before diving into the weeds, define notation for the real numbers;

$$\begin{aligned}\mathbb{R} &= (-\infty, \infty)\,,\\ \bar{\mathbb{R}} &= [-\infty, \infty]\,,\\ \mathbb{R}_+ &= [0, \infty)\,,\\ \bar{\mathbb{R}}_+ &= [0, \infty]\,,\end{aligned}$$

where $\bar{\cdot}$ represents the extended number system, which adds absolute minima and maxima to the set of real numbers. Define the Borel sigma-algebra as $\mathcal{B}(\bar{\mathbb{R}})$, which is the smallest sigma-algebra on the set of real-numbers.

Definition 2.7. *The mapping $f : E \to \bar{\mathbb{R}}$ is $\mathcal{E}$-measurable if and only if for every r in $\mathbb{R}$, $f^{-1}[-\infty, r] \in \mathcal{E}$.*

Proof. *This is just an application of Prop. 2.2. Let B be an element of a subset $\mathcal{F}_0$ of the real numbers. Then, for f to be measurable wrt $\mathcal{E}$, for each B, $f^{-1}B \in \mathbb{R}$.* ■

2.2.2 Positive and negative parts of a function.

A function f can be decomposed into negative and positive parts. Let the notation $a \wedge b$ and $a \vee b$ denote the minimum and maximum between a and b respectfully.

Let f be a numerical function on E. Then,

$$f^+ = f \vee 0, \quad f^- = -(f \wedge 0), \tag{2.20}$$

denote the positive and negative parts of the functions.

Definition 2.8. *The function f is $\mathcal{E}$-measurable, meaning that is is a map $f : E \to \mathbb{R}$ and is measurable from $\mathcal{E}$ to $\mathcal{B}(\mathbb{R})$ if and only if both f^+ and f^- are.*

Proof. *The decomposition of f is shown below;*

$$f = f^+ - f^-, \tag{2.21}$$

where,

$$f^+ = f \vee 0, \quad f^- = -(f \wedge 0) \tag{2.22}$$

To prove this, we need to show that the decomposition is measurable using Prop 2.3 in Cinlar.

For a pair $(E, \mathcal{E})$, the domain of both of these functions are elements of E, or some subset in $\mathcal{E}$. From this, it is obvious that when one of the functions has a non-zero image, the other part will be zero. It should be noted that the quantities f^+ and f^- are both greater than or zero to 0.

Now, the proof relies on checking for measurability. Let $\mathcal{E}_0$ be the subset of $\mathcal{E}$ which returns a non-zero value in f^+. This collection of subsets, $\mathcal{E}_0 \subset \mathcal{E}$. This set is a generator of $\sigma\mathcal{E}$. Thus, it is necessary to show measurability on this subset.

By construction of a numerical function, every element in $\mathcal{E}$ is measurable with respect to $\mathcal{B}(\mathbb{R})$.

In other words, if f is measurable, the functions f^+, f^- are measurable as the transformations

$$f^+ = \frac{|f| + f}{2}, f^- = \frac{|f| - f}{2}, \tag{2.23}$$

are continuous, which implies that f^+, f^- are measurable if f is measurable. The converse is an application of the composition of measurable functions. ■

2.2.3 Indicators and simple functions

Define the indicator as the function below;

$$1_A(x) = \begin{cases} 1 & \text{if} x \in A\,, \\ 0 & \text{if} x \notin A\,. \end{cases} \tag{2.24}$$

We can see that this function is $\mathcal{E}$-measurable if and only if $A \in E$. A function f is considered simple if it has the form;

$$f = \sum_{i=1}^{n} a_i 1_{A_i}\,, \tag{2.25}$$

where n is a natural number, a_i is a real number, and A_i is a measurable set.

2.2.4 Limits of sequences of functions

Let (f_n) be a sequence of numerical functions on E. The functions

$$\inf f_n, \quad \sup f_n, \quad \liminf f_n, \quad \limsup f_n \tag{2.26}$$

are defined on E point-wise. In general, if the function at x has a limit inferior equal to the limit superior, then the function is said to have a point-wise limit f and we write

$$f = \lim f_n\,. \tag{2.27}$$

Definition 2.9. *Let (f_n) be a sequence of $\mathcal{E}$-measurable functions. Then, each of the functions in Eqn. 2.26 is $\mathcal{E}$-measurable. Moreover, if it exists, $\lim f_n$ is $\mathcal{E}$-measurable.*

2.2.5 Approximation of measurable functions

For each $n \in \mathbb{N}^+$, let $d_n(r)$ denote the dyadic function:

$$d_n(r) = \sum_{k=1}^{n2^n} \frac{k-1}{2^n} 1_{\left[\frac{k-1}{2^n}, \frac{k}{2^n}\right)}(r) + n1_{[n,\infty]}(r)\,. \tag{2.28}$$

Naturally, this looks confusing, so we can plot a couple examples.

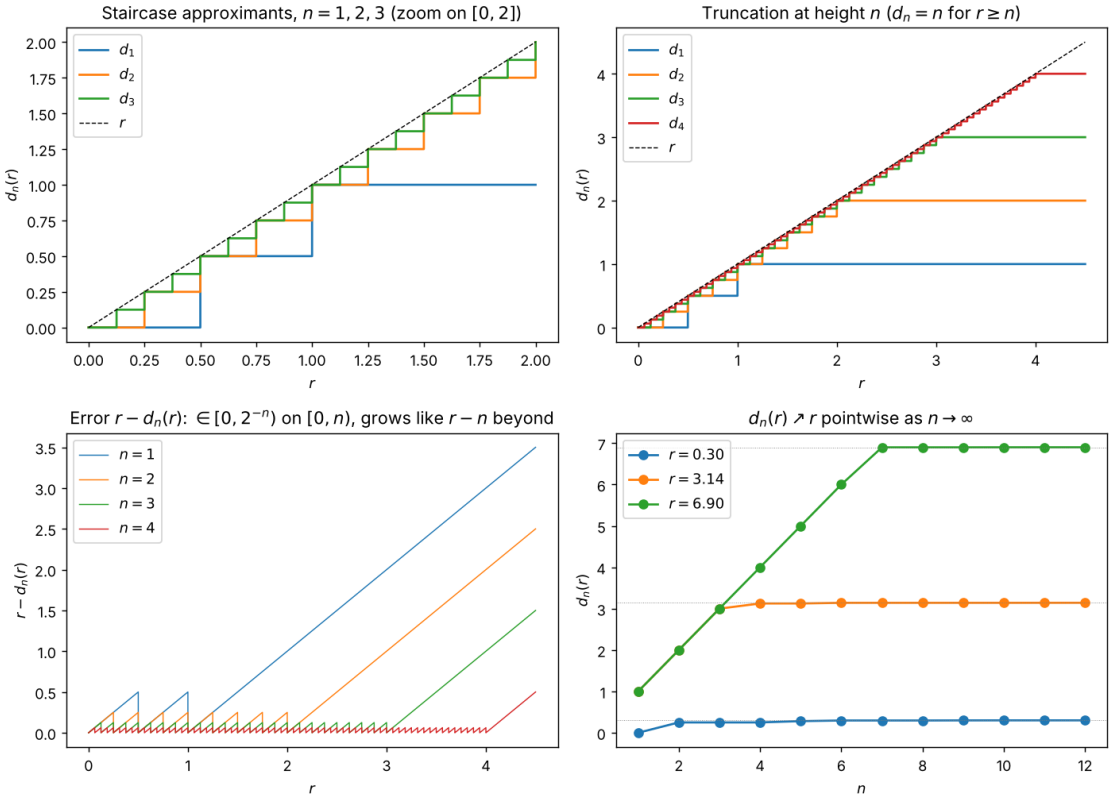


Figure 2.1: Dyadic functions drawn using code written by CLAUDE SONNET 5.5 (MEDIUM) in PYTHON. Panel 1 shows the approximations of the dyadic to the function r, which it is approximating. Panel 2 shows the truncation at the height of n. Panel.3 shows the error, as a function of r. Panel 4 then shows the point-wise convergence. It is clear as $n \to \infty$ the dyadic $d_n(r) \nearrow r$.

Using this function, a very useful and powerful theorem can be stated with reduces measurable functions to a set of simple functions and a limit.

Definition 2.10. *A positive function on E is $\mathcal{E}$-measurable if and only if it is the limit of an increasing sequence of positive simple functions.*

2.2.6 Problems

Problem 2.6. *Let E be a set and $(F, \mathcal{F})$ a measurable space. For $f : E \to F$, define*

$$f^{-1}\mathcal{F} = \{f^{-1}B : B \in \mathcal{F}\}$$

Where $f^{-1}B = \{x \in E : f(x) \in B\}$. Show that $f^{-1}\mathcal{F}$ is a sigma-algebra on E.

Solution. Since $f^{-1}\mathcal{F} = \{f^{-1}B : B \in \mathcal{F}\}$ we need to need to show the properties. Since $f^{-1}F = E$, $E \in f^{-1}\mathcal{F}$. Similarly $f^{-1}\emptyset = \emptyset$, $\emptyset \in f^{-1}\mathcal{F}$. To show complement, consider $A \in \mathcal{F}$.

Then $A^c \in \mathcal{F}$. Thus, we require by definition that $f^{-1}A^c \in E$. Thus the set is closed under complement for all $A \in \mathcal{F}$. For Union and intersection we use the definition;

$$\left(\bigcup_{i=1}^{\infty} A_i\right) \in \mathcal{F}$$
$$f^{-1}\left(\bigcup_{i=1}^{\infty} A_i\right) \in E$$

$$\left(\bigcap_{i=1}^{\infty} A_i\right)^c = \bigcup_{i=1}^{\infty} A_i^c$$
$$\in \mathcal{F}$$
$$f^{-1}\bigcap_{i=1}^{\infty} A_i \in E$$

. . . . . .

2.3 Measures

Let $(E, \mathcal{E})$ be a measurable space. A measure on $(E, \mathcal{E})$ is defined as a map $\mu : \mathcal{E} \to \bar{\mathbb{R}}_+$ where $\bar{\mathbb{R}}_+ = [0, +\infty]$ is the extended real number line.

The measure has properties. Let the sequence $A_1, A_2, \cdots \in \mathcal{E}$ be disjoint, meaning $A_i \cap A_j = \emptyset \forall i, j$;

$$\mu(\emptyset) = 0$$
$$\mu\left(\bigcup_n A_n\right) = \sum_n \mu\left(A_n\right),$$
$$\text{for every disjointed sequence } (A_n) \in \mathcal{E}.$$

The idea of the countable additivity can be reasoned by the logic of the Venn-diagram overlap. The area, or measure of these sets only add up together if there is no overlap between the sets.

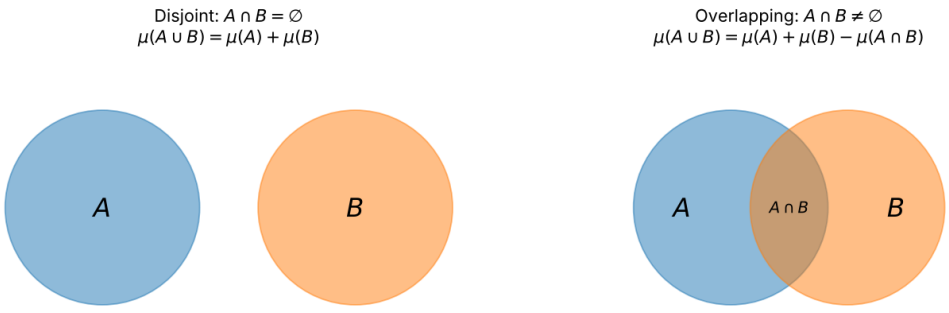


Figure 2.2: Venn-diagram of disjoint and non-disjoint sets.

The definition of the measure can be rephrased such that it contains several properties about the measure and the elements of the measure.

1. At least one set in $\mathcal{E}$ has finite measure. $\mu(A) = \mu(A \cup \emptyset) = \mu(A) + \mu(\emptyset) = \mu(A)$
2. If $A \cap B = \emptyset$ Then $\mu(A \cap B) = \mu(A) + \mu(B)$
3. If $A \subset B$ Then $\mu(A) \leq \mu(B)$
4. If $A_n \nearrow A$ Then $\mu(A_n) \nearrow \mu(A)$
5. $\mu\left(\bigcup_n A_n\right) \leq \sum_n \mu(A_n)$

Proof.

1. Def.
2. proof 1.
3. If $A \subset B$, then one can write that

$$B = (B \cap A) \cup (B \cap A^c)\,, \tag{2.29}$$

as shown in Fig. 1.1. Then, the measure of B is given by;

$$\begin{aligned}\mu(B) &= \mu((B \cap A) \cup (B \cap A^c)) \\ &= \mu(B \cap A) + \mu(B \cap A^c)\,.\end{aligned}$$

If $A \subset B$, then $B \cap A = A$;

$$\begin{aligned}&= \mu(B \cap A) + \mu(B \cap A^c)\,, \\ &= \mu(A) + \mu(B \cap A^c) \geq \mu(A)\,.\end{aligned}$$

Thus, $\mu(A) \leq \mu(B)$ if $A \subset B$.

4. proof 3
5. Expanding the LHS;

$$\begin{aligned}\mu\left(\bigcup_i^\infty A_i\right) &= \mu(A_1, \cup A_2, \dots) \\ &= \mu(A_1 \cup (A_2 \setminus A_1) \cup (A_3 \setminus (A_2 \setminus A_1)) \cup \dots) \\ \text{Disjoint sets} &= \mu(A_1) + \mu(A_2 \setminus A_1) + \mu(A_3 \setminus (A_2 \setminus A_1)) + \dots \\ &\leq \sum_i^\infty \mu(A_i)\,.\end{aligned}$$

which is trivial as the sets

■

Some examples of measures are listed below for the measurable space $(E, \mathcal{E})$;

1. Dirac Measure:

$$\delta_x(A) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}$$

2. Counting Measure: Let D be a fixed subset of E. The counting measure counts the number of points $A \cap D$. We interpret this as all elements outside D have zero measure and all elements of D have equal measure.

$$\#_D(A) = \sum_{x \in D} \delta_x(A), \quad A \in \mathcal{E}$$

3. Discrete Measure: Let D be a countable subset of E, and $m(x)$ a positive number.

$$\mu(A) = \sum_{x \in D} m(x)\delta_x(A), \quad A \in \mathcal{E}$$

4. Probability Measure: Let $(E, \mathcal{E})$ be a measurable space. A function $\mathbb{P} : \mathcal{E} \to [0, 1]$ that is a measure and has $\mathbb{P}(E) = 1$. The triplet $(E, \mathcal{E}, \mathbb{P})$ is known as a probability space. For ease, $\mathbb{P}(\cdot)$ is sometimes written as $p(\cdot)$.
5. Lebesgue Measure: The Lebesgue measure, $\lambda(\cdot)$ denotes the length of a subset of the Borel set. For example, if $E = \mathbb{R}$, then $\lambda((a, b)) = b - a$ if $b > a$.

A triplet $(E, \mathcal{E}, \mu)$ is known as a measure space.

2.3.1 Probability Measure

If $p : \mathcal{E} \to [0, 1]$ is a probability measure for any $A \in \mathcal{E}$. Then, by the countable additivity of the measure,

$$\begin{aligned} 1 = p(\Omega)\,, \\ &= p(A \cup A^c)\,, \\ &= p(A) + p(A^c)\,. \end{aligned}$$

which is the definition of the total probability.

2.3.2 Problems

. . . . . .

2.4 Integration

Let $(E, \mathcal{E}, \mu)$ be a measure space. Here we define $\mathcal{E}$ as the sigma algebra of all $\mathcal{E}$-measurable functions, which are all measurable functions mapping to the Borel sigma-algebra.

We define the 'integral of f with respect to μ' as:

$$\mu f = \mu(f) = \int_E \mu(dx) f(x) = \int_E f d\mu \tag{2.30}$$

Definition 2.11.

- *Let f be simple and positive, such that it has the form $f = \sum_{i=1}^n a_i 1_{A_i}(x)$. Then we define*

$$\mu f = \int \sum_{i=1}^n a_i 1_{A_i}(x) d\mu = \sum_{i=1}^n a_i \mu(A_i)$$

- *Let f be in the set of positive $\mathcal{E}$-measurable functions, denoted $\mathcal{E}^+$. Put $f_n = f(d_n(r))$ where $d_n(r)$ is the dyadic function approximating a numerical function. Then, each f_n is simple and positive, and the sequence (f_n) increases to f as per Thm. 2.10. The integral μf_n for each n is defined by item (a). Define the limit as*

$$\mu f = \lim \mu f_n \tag{2.31}$$

- *Let $f \in \mathcal{E}$. The functions f^+, f^- belong to $\mathcal{E}^+$ and the integrals $\mu(f^{+/-})$ are defined as per (b). Noting the functional decomposition, if at least one integral of these functions is finite, it exists. Otherwise, if $\mu(f^+) = \mu(f^-) = \infty$ then μf is undefined.*

This definition can be used on different measures to see how the integral is defined.

1. Dirac Measure:
$$\mu f = \delta_x(A) f(x) = f(A)$$

2. Discrete Measure:
$$\mu f = \sum_{x \in D} m(x) f(x)$$

3. Lebesgue Integrals:
$$\mu f = \text{Leb}_E f = \int_E \text{Leb}_E(dx) f(x) = \int_E dx f(x)$$
Note that if the lebesgue exists, so does the riemann integral, which is reminded by the notation in the last equality. Note that the converse is false.

The main idea of the integral with respect to a measure is packaged in how the measure acts on infinitesimal pieces dx. Since the measure gives different weights to different parts of the set. We can use a result that approximates a measurable function f as a convergent sequence of simple functions. That is, we can approximate f using simple functions that partition the domain in a way that converges to the value of the integral.

2.4.1 Integrability

A function is f in $\mathcal{E}$ is said to be integrable if μf exists and is a real number. Thus, f in $\mathcal{E}$ is integrable if and only if $\mu f^+ < \infty$ and $\mu f^- < \infty$. Additionally, the absolute value, $\mu|f| < \infty$.

Definition 2.12. *Every integrable function is real-valued almost everywhere.*

Proof. *?.* ■

2.4.2 Monotone Convergence Theorem

Definition 2.13. *Let (f_n) be an increasing sequence in $\mathcal{E}^+$. Then,*
$$\mu(\lim f_n) = \lim \mu f_n \tag{2.32}$$

Proof. *Let $f = \lim f_n$. Since (f_n) is increasing, the integrals μf_n form an increasing sequence of numbers by the monotonicity property, thus $\lim \mu f_n$ exists. Since $f \geq f_n$ for all n, $\mu f \geq \mu f_n$.*

The reverse inequality is outlined in Sec 4 of Erlan Cinlar. ■

2.5 Transforms and Indefinite Integrals

2.5.1 Image measures

Let $(F, \mathcal{F})$, $(E, \mathcal{E})$ be measurable spaces. Let ν be a measure on $(F, \mathcal{F})$. Let the function $h : F \to E$. The mapping $\nu \circ h^{-1}$ maps from the sigma-algebra $\mathcal{E}$ into $\mathbb{R}^+$.
$$\nu(h^{-1}(B)), \quad B \in \mathcal{E}\,. \tag{2.33}$$

To see this, h^{-1} is the pre-image of h, which maps from $E \to F$. The measure on F maps from F to the real numbers. Here, the composition function is known as the image of ν under h.

2.5.2 Radon-Nikodym Theorem

Given a measurable space $(X, \mathcal{X})$, suppose μ and ν are σ-finite measures. If ν is absolutely continuous with respect to μ (written as $\nu \ll \mu$), meaning $\mu(E) = 0 \implies \nu(E) = 0$ for all measurable sets E, then there exists a measurable function $f : X \to [0, \infty)$ such that:
$$\nu(E) = \int_E f \, d\mu \quad \text{for all } E \in \mathcal{X} \tag{2.34}$$

Chapter 3

Probability

This section is written from my notes while enrolled in UBC's STAT 560 course on statistical inference and probability. These notes are supplemented by any further mentioned books. Notably, Erlan Cinlar's "Probability and Stochastics."

3.1 The Probability Measure

Let Ω be defined as the sample space. The set Ω is measurable if for any subset A and B of Ω the properties:

$$A \cup B \in \Omega\,,$$
$$A \cap B \in \Omega\,,$$
$$A \setminus B \in \Omega\,.$$

The collection, σ_Ω, of subsets of Ω is called a sigma-algebra on Ω if it is closed under finite union and complement. This is opposed to a collection O which may not be necessarily closed under union and complement but can still constitute an algebra. The pair, (Ω, σ_Ω), is a measurable-space when σ_Ω is a sigma-algebra. Given a measurable-space, a measure, μ, can be defined as a map $\mu : \sigma_\Omega \to \mathbb{R}^+$, such that $\mu(\emptyset) = 0$ and $\mu(\bigcup_n A_n) = \sum_n \mu(A_i)$.

Definition 3.1. *The probability measure on a measurable-space (Ω, σ_Ω), $\mathbb{P}$ is defined by the following properties. Let H and K be events, or elements of Ω:*

$$\mathbb{P}(\emptyset) = 0, \quad \mathbb{P}(\Omega) = 1\,, \tag{3.1}$$
$$H \in K \to \mathbb{P}(H) < \mathbb{P}(K)\,, \tag{3.2}$$
$$H \cap K = \emptyset \to \mathbb{P}(H \cup K) = \mathbb{P}(H) + \mathbb{P}(K) \tag{3.3}$$
$$\textit{if } H_n \textit{ is disjoint}, \quad \mathbb{P}(\bigcup_n H_n) = \sum_n \mathbb{P}(H_n) \tag{3.4}$$
$$\mathbb{P}(\bigcup_n H_n) \leq \sum_n \mathbb{P}(H_n) \tag{3.5}$$

For the purposes of applied statistics, one can simplify these properties into a set of axioms with no real issues. Let the space $(\Omega, \sigma_\Omega, \mathbb{P})$ be a probability space, then the following properties are true;

1. $\mathbb{P}(A) \geq 0 \forall A \in \{A\}$
2. $\mathbb{P}(\Omega) = 1$.
3. if A_n are disjoint, then $\mathbb{P}(\bigcup_n A_n) \leq \sum_n \mathbb{P}(A_n)$.

3.1.1 Random Variables

Given a probability space $(\Omega, \mathcal{H}, p)$ and a measurable space $(E, \mathcal{E})$, a random variable, X, is defined as a map $X : \Omega \to E$ such that it is measurable relative to $\mathcal{H}$ and $\mathcal{E}$. That is if $X^{-1}A$ is an event in $\mathcal{E}$, then

$$X^{-1}A = \{X \in A\} = \{\omega \in \Omega : X(\omega) \in A\}\,. \tag{3.6}$$

What this means is that $X^{-1}A$ is the set of elements in the sample space such that X acting on that set is in the event. Mathematically, this implies that the pre-image of A under X is contained within the sigma-algebra of the probability space, $\mathcal{H}$.

The distribution of a random variable is denoted by the image measure of p. Here, let μ denote the image measure of p under X;

$$\mu(A) = p(X^{-1}A) = p(X \in A), \quad A \in \mathcal{E}\,. \tag{3.7}$$

Here, μ is a probability measure on $(E, \mathcal{E})$ and is called the distribution of X. We need to show that if μ and some other measure is finite and agree[1] on a p-system $\mathcal{C}$ on E, then they are identical. Defining a function $c : \mathbb{R} \to [0, 1]$ we can specify,

$$c(x) = \mu[-\infty, x] = \mathbb{P}(X \leq x), x \in \mathbb{R}\,. \tag{3.8}$$

It should be noted that in traditional probability theory, these functions are very useful as they allow one to avoid the use of measures.

3.1.2 Examples of Distributions

1. Poisson Distribution;

$$\mathbb{P}(x = n) = \frac{\lambda^n}{n!}e^{-\lambda}, \quad n \in \mathbb{N}\,. \tag{3.9}$$

   Alternatively, one can specify the corresponding distribution's probability measure, μ.

$$\mu(A) = \sum_{n \in A} \frac{\lambda^n}{n!}e^{-\lambda}, A \subset \mathbb{N}\,. \tag{3.10}$$

2. Exponential Distribution;

$$\mu(dx) = dxce^{-cx}\,. \tag{3.11}$$

[1] For every single set A in the p-system $\mathcal{A}$, the two measures have the same numerical value.

3.2 Multivariate Distributions

Let $\mathbf{X}$ be a random vector with components $X_1, \dots X_n$. The distribution is arrives from is $P(x_1, x_2, \dots x_n)$.

The marginal distribution is defined as

$$P(x_1, \dots, x_s) = \int P(x_1, \dots, x_s, x_{s+1}, \dots, x_r) dx_{s+1} \dots dx_r$$

where the left hand side is defined as the conditional distribution. It is often written in terms of Baye's rule, as well as the set defined probability;

$$P(x_1, \dots, x_s | x_{s+1} \dots x_r) = \frac{P(x_1, \dots, x_s, x_{s+1}, \dots, x_r)}{P(x_{s+1}, \dots, x_r)} \tag{3.12}$$

And written in set notation;

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \tag{3.13}$$

The moments of the multivariate distribution are defined as

$$\left\langle \prod_{i=1} X_i^{m_i} \right\rangle = \int x_1^{m_1} x_2^{m_1} \dots x_n^{m_n} P(x_1, x_2, \dots, x_n) dx_1 dx_2, \dots dx_n$$

The characteristic function occurs as the product of Fourier transforms;

$$\begin{aligned} G(k_1, \dots, k_n) &= \left\langle e^{i(k_1 X_1 + \dots + k_n X_n)} \right\rangle \\ &= \sum_{i=0}^{\infty} \frac{(ik_1)^{m_1} (ik_2)^{m_2} \dots (ik_n)^{m_n}}{m_1! m_2! \dots m_n!} \left\langle \prod_{i=1} X_i^{m_i} \right\rangle \end{aligned}$$

Where one extracts the coefficients from the expansion. The log of this function extracts the cumulants, known as the cumulant generating function;

$$\log[G(k_1, \dots, k_n)] = \sum_{i=0}^{\infty} \frac{(ik_1)^{m_1} (ik_2)^{m_2} \dots (ik_n)^{m_n}}{m_1! m_2! \dots m_n!} \left\langle \prod_{i=1} X_i^{m_i} \right\rangle_c \tag{3.14}$$

3.3 Covariance Matrix

The covariance matrix is defined as the $n \times n$ matrix $\langle X_i X_j \rangle_c$ Which stands for the second cumulant, or the variance.

$$\langle X_i X_j \rangle_c = \langle X_i X_j \rangle - \langle X_i \rangle \langle X_j \rangle \tag{3.15}$$

When normalized, this matrix forms the correlation coefficients between the random variables:

$$\rho_{i,j} = \frac{\langle X_i X_j \rangle - \langle X_i \rangle \langle X_j \rangle}{\sqrt{(\langle X_i^2 \rangle - \langle X_i \rangle^2)(\langle X_j^2 \rangle - \langle X_j \rangle^2)}} \tag{3.16}$$

3.3.1 Transformations

Transformations of random-variables answer the question of given X with PMF/PDF f_X, how does one calculate the distribution of some $Y = r(X)$? The ben-br notes define a simple process;

Theorem 3.1. *The transformation of a random variable can be given by the following procedure;*

- *For each y, find the set $A_y = \{x : r(x) \leq y\}$*
- *Find the CDF of y*

$$F_Y(y) = \int_{A_y} f_X(x) dx \tag{3.17}$$

- *The PDF is the derivative of the CDF;*

$$f_Y(y) = F'(y) \tag{3.18}$$

For transformations that are strictly monotone, then there is a well-defined inverse, $s = r^{-1}$, then the process is a simple formula;

$$f_Y(y) = f_X(s(y))\left|\frac{ds(y)}{dy}\right| \tag{3.19}$$

Example 3.1. *Let $X \sim Exp(\lambda)$ and $r = \xi^2$. Then how can we find the transformation $Y = r(X)$?*

1. *Using the general method, we can calculate the PDF exactly;*

$$\begin{aligned} p(Y \leq y) &= p(r(X) \leq y) \\ &= p(X^2 \leq y) \\ &= p(X \leq \sqrt{y}) \\ F_Y(y) &= 1 - e^{-\lambda\sqrt{y}} \\ f_Y(y) &= \frac{\lambda}{2\sqrt{y}} e^{-\lambda\sqrt{y}} \end{aligned}$$

2. *Since $r(X)$ is a monotonic function with an inverse $s(x) = \sqrt{x}$ then we can compute the formula directly;*

$$\begin{aligned} f_Y(y) &= \lambda e^{-\lambda\sqrt{y}} \cdot \left|\frac{1}{2\sqrt{y}}\right| \\ &= \frac{\lambda}{2\sqrt{y}} e^{-\lambda\sqrt{y}} \end{aligned}$$

Thus, both methods yield the correct transformed PDF.

3.3.2 Expectations

The expectation of a random variable is given by the explicit integral over the CDF. Specifically, we are interested in the value that is most likely to occur. This can be formalized by adding all of the cumulative probability over the range of values. Here we assume that X is non-negative, which eases the bounds of integration. Otherwise, this can be recast to have a negative sign to ensure all regions are positive;

$$\int_0^\infty p(X \geq x)dx = \int_0^\infty \left(\int_x^\infty f_Y(y)dy\right) dx$$

This integral sweeps a region in xy space. This region is fixing x from $0 \to \infty$. Then, y integrates from x to infinity;

$$0 \leq x, \quad x \leq y \leq \infty$$

Thus, we can integrate instead over x which is from 0 to y, then overall all space;

$$\begin{aligned} \int_0^\infty \left(\int_x^\infty f_Y(y)dy\right) dx &= \int_0^\infty \left(\int_0^y f_Y(y)dx\right) dy \\ &= \int_0^\infty y f_Y(y)dy \\ &:= \mathbb{E}[y] \end{aligned}$$

Thus, the expectation is defined more commonly by the law of the unconscious statistician;

Theorem 3.2. *Expectation of random variable or function of a random variable is defined by a CDF of a random variable;*

$$\mathbb{E}\left[r(x)\right] = \int_\Omega r(x) f_X(x)dx \tag{3.20}$$

3.3.3 Characteristic Functions

3.4 Problems

Problem 3.1. *Given N people in a room, what is the probability that 2 or more people share a birthday?*

Solution. This problem is an exercise in counting. First, lets go case-by-case and write out the general solution. There are 365 days in total. If there is only 1 person in the room, only 1 day is taken. Once we place the second person in the room, we can calculate the probability of the two individuals not-sharing the same birthday;

$$\mathbb{P}[\textbf{share birthday}] = 1 - \mathbb{P}[\textbf{ not share birthday}] \tag{3.21}$$

Ie, the second person has only 364 days to choose from to not share. Thus, their probability of not sharing a birthday are 364/365, ie the event space over the sample space. Likewise, the following person has now only 363 days to choose from and so on. This relation can be multiplied together such that the overall probability of not sharing is that nobody in the room is sharing. This can be built using a product;

$$\mathbb{P}[\textbf{not share birthday}] = \prod_{x=1}^{N} \frac{365 - x}{365} \tag{3.22}$$

This gives us the probability that at least 2 people are sharing;

$$\mathbb{P}[\textbf{at least 2 share birthday}] = 1 - \prod_{x=1}^{N} \frac{365 - x}{365} \tag{3.23}$$

Plotting this relation as a function of N, we can see the shape of this relation in Fig. 3.1.

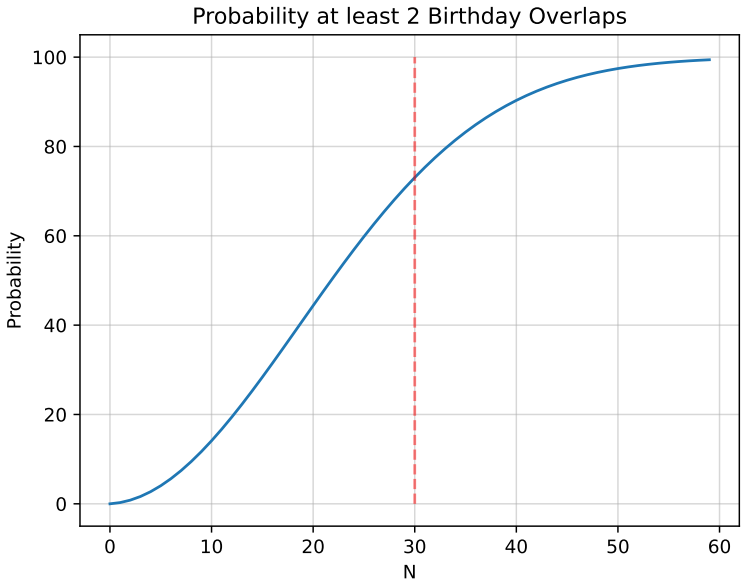


Figure 3.1: Probability of at least 2 people in a room sharing a birthday as a function of N people in the room. at $N = 30, \mathbb{P} \approx 73\%$

Chapter 4

Stochastic Processes

4.1 Point Processes

Define a point process by the following rules. The sample space is consisting of a set of states which consist of $s = 0, 1, 2, \ldots$. For each s, there is a set of s real numbers τ which is increasing.

The probability distribution of these states is given by a sequence of non-negative functions Q_s. The normalization is given by;

$$1 = Q_0 + \int_{-\infty}^{\infty} d\tau_1 Q_1(\tau_1) + \int_{-\infty}^{\infty} d\tau_1 \int_{-\infty}^{\infty} d\tau_2 Q_2(\tau_1, \tau_2) + \ldots \tag{4.1}$$

It is common to permit only symmetric functions of $Q_s(t_1, \ldots, t_s)$ such that there are $s!$ permutations for each Q_s. Then the normalization condition becomes simple to write as a series;

$$1 = Q_0 + \sum_{s=1}^{\infty} \frac{1}{s!} \int_{-\infty}^{\infty} Q_s(t_1, \ldots, t_s) dt_1 \cdot \cdots \cdot dt_s \tag{4.2}$$

Here it appears that $Q_s(t_1, \ldots, t_s)$ is known as the probability of the sequence of events where s dots occur at some vector t_1. Here, t can denote a time in a temporal process, as well as spatial coordinates.

For example, consider an ideal gas. The ideal gas has a volume in $\mathbb{R}^3$. Such a system can be defined by the unique property of each state, which in this case, can be the energy of the system. The number of discrete microstates is given by the relation to entropy

$$S = k_b \ln \Omega$$

Thus, we can find the number of microstates

$$\Omega = e^{S/k_b}$$

the probability of a single state, which consists of a state variable is proportional by definition of probability;

$$P(N, U, P, V) = Q_s(U) = \frac{1}{\Omega} = e^{-S/k_b} = e^{-\beta(U + PV - \mu N)} \tag{4.3}$$

This is defined as the Janossy density for a single state. The normalization condition is

$$\begin{aligned}
1 &= \sum_{n=0}^{N} e^{-\beta(U+PV-\mu n)} \\
&= e^{-\beta(U+PV)} \sum_{n=0}^{N} e^{\beta\mu n} \\
&= e^{-\beta(U+PV)} \frac{1-e^{\beta\mu N}}{1-e^{\beta\mu}}
\end{aligned}$$

Another example is for a general counting process. Suppose that $Q_s(t_1, \ldots, t_s)$ is the probability of exactly s detections at intervals dt_s. We can compute the average number of dots by defining

$$N(t) = \sum_{j=1}^{s} \mathbf{1}_{t_{j-1}<t<t_j}$$

. This just counts the number of dots by summing each interval by an indicator function.

$$\begin{aligned}
\langle N(t) \rangle &= \sum_{s=0}^{\infty} \frac{1}{s!} \int_{-\infty}^{\infty} dt_1 \ldots dt_s Q_s(t_1, \ldots, t_s) \sum_{j=1}^{s} \mathbf{1}_{t_{j-1}<t<t_j} \\
s \text{ symmetric integrals over } [t_a, t_b] &= \sum_{s=1}^{\infty} \frac{s}{s!} \int_{t_a}^{t_b} dt_1 \int_{-\infty}^{\infty} \ldots dt_s Q_s(t_1, \ldots, t_s) \\
&= \sum_{s=1}^{\infty} \frac{1}{(s-1)!} \int_{t_a}^{t_b} dt_1 \int_{-\infty}^{\infty} \ldots dt_s Q_s(t_1, \ldots, t_s)
\end{aligned}$$

And similarly for the second moment,

$$\begin{aligned}
\langle N(t)^2 \rangle &= \sum_{s=0}^{\infty} \frac{1}{s!} \int_{-\infty}^{\infty} dt_1 \ldots dt_s Q_s(t_1, \ldots, t_s) \left[\sum_{i=1}^{s} \mathbf{1}_{t_{i-1}<t<t_i} \right] \left[\sum_{j=1}^{s} \mathbf{1}_{t_{j-1}<t<t_j} \right] \\
&= \sum_{s=0}^{\infty} \frac{1}{s!} \int_{-\infty}^{\infty} dt_1 \ldots dt_s Q_s(t_1, \ldots, t_s) \left[\sum_{i=j} \mathbf{1}_{t_{j-1}<t<t_j} + \sum_{i \neq j} \mathbf{1}_{t_{j-1}<t<t_j} \right] \\
s \text{ cases where } (i=j) \to &= \langle N \rangle + \sum_{s=0}^{\infty} \frac{1}{s!} \int_{-\infty}^{\infty} dt_1 \ldots dt_s Q_s(t_1, \ldots, t_s) \left[\sum_{i \neq j} \mathbf{1}_{t_{j-1}<t<t_j} \right] \\
s \text{ ways to fix } i, (s-1) \text{ fix } j &= \langle N \rangle + \sum_{s=2}^{\infty} \frac{s(s-1)}{s!} \int_{t_a}^{t_b} dt_1 \int_{t_a}^{t_b} dt_2 \int_{-\infty}^{\infty} dt_3 \ldots dt_s Q_s(t_1, \ldots, t_s) \\
&= \langle N \rangle + \sum_{s=2}^{\infty} \frac{1}{(s-2)!} \int_{t_a}^{t_b} dt_1 \int_{t_a}^{t_b} dt_2 \int_{-\infty}^{\infty} dt_3 \ldots dt_s Q_s(t_1, \ldots, t_s)
\end{aligned}$$

Thus, we have recovered the variance of a general process being that of a Poisson process;

$$\langle N \rangle = \langle N^2 \rangle - \langle N \rangle^2$$

This is suprising as we haven't said anything about how Q_s is distributed!

4.1.1 Poisson processes

Random dot events are independent if the Janossy Density factorizes into the form of an independent joint probability density;

$$Q_s(t_1, \ldots, t_s) = e^{-\nu} p(\tau_1) \ldots p(\tau_s) \tag{4.4}$$

In this case, one can see that if we define

$$\langle N \rangle = \nu = \int_{t_a}^{t_b} p(\tau) d\tau$$

we find the following result.

$$\begin{aligned}
\langle e^{ikN} \rangle &= e^{-\nu} \sum_{s=1}^{\infty} \frac{1}{s!} \left[\int_{\infty}^{\infty} e^{ik \sum_\sigma 1_{t_\sigma}} p(\tau) d\tau \right]^s \\
&= \exp \left[\int_{\infty}^{\infty} e^{ik \sum_\sigma 1_{t_\sigma}} p(\tau) d\tau - \nu \right] \\
&= \exp \left[\int_{\infty}^{\infty} \left(e^{ik \sum_\sigma 1_{t_\sigma}} - 1 \right) p(\tau) d\tau \right] \\
\text{surviving terms are only in } [t_a, t_b] &= \exp \left[\left(e^{ik} - 1 \right) \int_{t_a}^{t_b} p(\tau) d\tau \right]
\end{aligned}$$

This characteristic function is precisely that of the Poisson distribution, notably that the probability of N dots falling in an interval $[t_a, t_b]$ is;

$$P(N) = \frac{\langle N \rangle^N}{N!} e^{-\langle N \rangle} \tag{4.5}$$

This result is that an independent waiting time, or an independent process probability gives rise to a Poisson distributed process.

4.2 Stochastic Variables and Properties

As known, we can define a stochastic process for a given random variable and some function. Typically, if this is a value of time, it is known as a general stochastic process.

$$Y_X(t) = f(X, t) \tag{4.6}$$

For example, assume the form $Y(t) = \psi(t)X$. Assume that here $\psi(t) > 0$ such that the math works out nicely, otherwise, the probability inequality will switch signs.

$$Y(t) = \psi(t)X \tag{4.7}$$

If we know the distribution of X, one can write that

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(x) dx \tag{4.8}$$

Thus, we can calculate the CDF of the transformed random variable;

$$\begin{aligned}
P(Y \leq y) &= P(\psi(t)X \leq y) \\
&= F_X \left(\frac{1}{\psi(t)} y \right), \quad \psi(t) > 0
\end{aligned}$$

The requirement on $\psi(t)$ is a valid one, as if the function is ever equal to 0, then the process is not random as it is simply 0 for that time. We can then calculate the PDF of Y;

$$\begin{aligned} f_Y(y) &= \frac{d}{dy} F_X\left(\frac{1}{\psi(t)} y\right) \\ &= \frac{1}{|\psi(t)|} f_X\left(\frac{1}{\psi(t)} y\right) \end{aligned}$$

Expectations

We can take expectations as normal on the random variable. It is interesting to see how the expectations collapse;

$$\langle Y(t) \rangle = \int_{-\infty}^{\infty} Y_x(t) P_X(x) dx \tag{4.9}$$

For the above example, we find a the same solution using the definition of the underlying distribution and from the transformed one;

$$\begin{aligned} \langle Y(t) \rangle &= \psi(t) \int_{-\infty}^{\infty} x f_X(x) dx \\ &= \int_{-\infty}^{\infty} y f_Y(y) dy \\ &= \int_{-\infty}^{\infty} \psi(t) x \frac{1}{|\psi(t)|} f_X\left(\frac{1}{\psi(t)} y\right) dy \\ &= \int_{-\infty}^{\infty} x f_X\left(x\right) \psi(t) dx \\ &= \psi(t) \int_{-\infty}^{\infty} x f_X(x) dx \end{aligned}$$

Autocorrelation

As a reminder, the auto-correlation is defined as

$$\kappa(i,j) = \langle (X_i - \langle X_i \rangle)(X_j - \langle X_j \rangle) \rangle \tag{4.10}$$

so, for a stochastic process, the function is a function of two time points;

$$\begin{aligned} \kappa(t_1, t_2) &= \langle (Y(t_1) - \langle Y(t_1) \rangle)(Y(t_2) - \langle Y(t_2) \rangle) \rangle \\ &= \langle Y(t_1) Y(t_2) \rangle - \langle Y(t_2) \rangle \langle Y(t_1) \rangle \end{aligned}$$

Once again, for the separable example above, we see that the time dependent functions factor out;

$$\kappa(t_1, t_2) = \psi(t_1)\psi(t_2) \left[\left\langle x^2 \right\rangle - \langle x \rangle^2 \right]$$

4.3 Stationary Processes

As we write earlier, a general stochastic process Y depends on some variables and some random variables. A stationary stochastic processes is one that has a stationary, or non-unique varinace and mean along the temporal variable. Typically, these are reserved to contain meand and variances that coincide with the time-step between time.

For example, consider two processes; the Wiener process, and the Ornstein-Uhlenbeck process.

The Wiener process is defined as the following,

$$W_t - W_s \sim \mathcal{N}(0, t - s)$$

probabilistically, this is describing the probability of moving to some W_t given some W_s. We can fully defined this process by defining the transition probability, as well as the initial distributive probability;

$$P_1(y_1, t) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{y_1^2}{2t}}$$
$$P_{1|1}(y_2, t_2|y_1, t_1) = \frac{1}{\sqrt{2\pi(t-s)}} e^{-\frac{(y_2-y_1)^2}{2(t-s)}}$$

This is compared to the Ornstein-Uhlenbeck process, which is notably defined as Markov and Stationary, with an autocorrelation of $e^{-\tau}$. Let $\tau = t - s$;

$$P_1(y_1, t) = \frac{1}{\sqrt{2\pi}} e^{-\frac{y_1^2}{2}}$$
$$P_{1|1}(y_2, t_2|y_1, t_1) = \frac{1}{\sqrt{2\pi(1-e^{-2\tau})}} e^{-\frac{(y_2-y_1)^2}{2(1-e^{-2\tau})}}$$

As one can see, the OH process is stationary as it forgets its initial position, whilst the Wiener process has a memory of this, hence, it is not markovian.

4.4 Power Spectra

4.5 Probabilisitic Hierarchy

The probabilistic Hierarchy is defined as a way to form a joint distribution for a stochastic process. Specifically, we can write down the following statement;

$$P(y_1, t_1; y_2, t_2; \ldots; y_n, t_n) = \int \delta(y_1 - Y_x(t_1))\delta(y_2 - Y_x(t_2)) \ldots \delta(y_n - Y_x(t_n)) p_X(x) dx \tag{4.11}$$

Here, $Y_x(t)$ is a realization of a process. We are then integrating over this realization weighted by the probability of each random value.

Example 4.1. *Suppose that $Y_x(t) = X\cos(\omega t)$ for $X \sim \mathcal{N}(0,1)$. Then, we can find $P(y_1, t_1)$*

$$\begin{aligned} P(y_1, t_1) &= \int \delta(y_1 - Y_x(t_1)) P_X(x) dx \\ &= \int \delta(y_1 - x\cos(\omega t)) \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \end{aligned}$$

Using the fact that

$$\delta(y_1 - x\cos(\omega t)) = \frac{1}{\cos(\omega t)} \delta(x - \frac{y}{\cos(\omega t)})$$

o

$$\begin{aligned} &= \int \delta(y_1 - x\cos(\omega t)) \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \\ &= \frac{1}{\cos(\omega t)} \int \delta(x - \frac{y_1}{\cos(\omega t)}) \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \\ &= \frac{1}{\cos(\omega t)} \frac{1}{\sqrt{2\pi}} e^{-\left(\frac{y_1}{\cos(\omega t)}\right)^2/2} \end{aligned}$$

Thus,

$$\boxed{P(y_1, t_1) = \frac{1}{\cos(\omega t)} \frac{1}{\sqrt{2\pi}} e^{-\frac{(y_1)^2}{2\cos^2(\omega t)}}} \tag{4.12}$$

4.6 Markov Processes

Markov processes are defined by a set of probabilistic equations.

4.7 Chapman-Kolmogorov Equations

4.8 Problems

Problem 4.1. *Find the mean-squared distance after N steps on a random walk on a 2D square lattice when U-turns are forbidden.*

Solution. The transition probability of this problem requires we know of the current and previous locations. Let n be the (x, y) position at some time, and m be the position at the previous time. Let $\xi_1 = n_1 - m_1$ be the difference of the two times. Since the steps have equal probability, we can write the transition matrix of this problem. Let $\uparrow, \downarrow, \rightarrow, \leftarrow$ denote the vectors of length 1 moving in the respective directions.

$$\begin{aligned} T(\xi_2|\xi_1) = &\frac{\delta(\xi_1 - \rightarrow)}{3}\left[\delta(\xi_2 - \leftarrow) + \delta(\xi_2 - \uparrow) + \delta(\xi_2 - \downarrow)\right] \\ &+ \frac{\delta(\xi_1 - \leftarrow)}{3}\left[\delta(\xi_2 - \rightarrow) + \delta(\xi_2 - \uparrow) + \delta(\xi_2 - \downarrow)\right] \\ &+ \frac{\delta(\xi_1 - \uparrow)}{3}\left[\delta(\xi_2 - \rightarrow) + \delta(\xi_2 - \leftarrow) + \delta(\xi_2 - \downarrow)\right] \\ &+ \frac{\delta(\xi_1 - \downarrow)}{3}\left[\delta(\xi_2 - \rightarrow) + \delta(\xi_2 - \uparrow) + \delta(\xi_2 - \leftarrow)\right] \end{aligned} \tag{4.13}$$

Given this transition, we can calculate the either the characteristic function or moments explicitly. Since there aren't that many delta functions to calculate, we can directly calculate the moments. From the defintion of ξ, we have that;

$$\vec{x}_n = \vec{x_0} + \sum_{j=1}^{N} \vec{\xi}_j \tag{4.14}$$

Thus, assume that $x_0 = 0$ and we can compute moments around $\vec{x}$ instead of ξ.

$$\begin{aligned} \left\langle \vec{x}^2 \right\rangle &= \left\langle \left[\sum_{j=1}^{N} \vec{\xi}_j \right]^2 \right\rangle \\ &= \left\langle \left[\sum_{j=1}^{N} \vec{\xi}_j \right] \left[\sum_{i=1}^{N} \vec{\xi}_i \right] \right\rangle \\ &= \left\langle \sum_{i=1}^{N} \sum_{j=1}^{N} \vec{\xi}_i \vec{\xi}_j \right\rangle \\ &= \sum_{i=1}^{N} \left\langle \vec{\xi}_i^{\,2} \right\rangle + 2 \sum_{i=1}^{N} \sum_{j=1}^{i-1} \left\langle \vec{\xi}_i \vec{\xi}_j \right\rangle \end{aligned} \tag{4.15}$$

We compute the moments explicitly. Note that $\rightarrow + \leftarrow = 0; \uparrow + \downarrow = 0;$

$$\begin{aligned} \langle \xi \rangle &= \sum_{\xi_1} \left[\sum_{\xi_2} \xi_2 T(\xi_2|\xi_1) \right] P(\xi_1) \\ &= \frac{1}{4}\left(\frac{1}{3}(\uparrow + \rightarrow + \leftarrow) + \frac{1}{3}(\downarrow + \rightarrow + \leftarrow) + \frac{1}{3}(\uparrow + \downarrow + \leftarrow) + \frac{1}{3}(\uparrow + \rightarrow + \downarrow) \right) \\ &= 0 \end{aligned}$$

This being said, we can compute the square moment and the correlation in time. It can be easily shown that since $\rightarrow \cdot \rightarrow^T = 1$, then $\langle \xi_i^2 \rangle = 1$. The correlation can be computed for jointwise distance $|i-j|$. We will use the recursive property to find the correlation.

$$\begin{aligned}\langle \xi_{j+1} | \xi_j \rangle &= \frac{1}{3}\left(\xi_j + \perp + (-\perp)\right) \\ &= \frac{1}{3}\xi_j\end{aligned}$$

Thus, from the recursive property of the correlation;

$$\langle \xi_{j+k} \xi_j \rangle = 3^{-k} \tag{4.16}$$

Then, the MSD is fully calculated;

$$\begin{aligned}\left\langle \vec{x^2} \right\rangle &= \sum_{i=1}^{N} \left\langle \vec{\xi_i}^2 \right\rangle + 2\sum_{i=1}^{N}\sum_{j=1}^{i-1} \left\langle \vec{\xi_i}\vec{\xi_j} \right\rangle \\ &= N + 2\sum_{i=1}^{N}\sum_{j=1}^{i-1} \lambda^{i-j} \\ &= N + 2\lambda \sum_{k=0}^{N-1} (N-k)\lambda^k \\ &= N + 2\lambda \left[\frac{(N-1)(1-\lambda^N)}{1-\lambda} - \frac{\lambda - (N-1)\lambda^N + (N-1)\lambda^{N+1}}{(1-\lambda)^2} \right] \\ &= N + 2\lambda \left[\frac{N}{1-\lambda} - \frac{(1-\lambda^N)}{(1-\lambda)^2} \right] \\ &= \frac{N(1+\lambda)}{1-\lambda} - \frac{2\lambda(1-\lambda^N)}{(1-\lambda)^2}\end{aligned}$$

Thus, we have that the MSD after N steps is the equation above, for $\lambda = 1/3$;

$$\left\langle \vec{x^2} \right\rangle = \frac{N(1+\lambda)}{1-\lambda} - \frac{2\lambda^2(1-\lambda^N)}{(1-\lambda)^2} \tag{4.17}$$

Note that since $N(1+\lambda)/(1-\lambda) = 2N$ There is a higher order of diffusivity compared to the free case, which N.

Chapter 5

Electromagnetism

5.1 Debye-Huckel Theory

Debye-Huckel theory is a description of the electromagnetic interactions between ions in a solution of ionic compounds.

5.1.1 Poisson-Boltzmann

The potential in a field of ions must be solved in terms of electrostatics in a media. This is described by Poisson's equation

$$\nabla^2\Psi(\mathbf{r}) = -4\pi\frac{\rho(\mathbf{r})}{\varepsilon} \tag{5.1}$$

In a sea of ions, we must carefully define the charge density as it pertains to the system. In a general solution of ions, there will be j species of ions.

$$\nabla^2\Psi(\mathbf{r}) = -4\pi\frac{\rho_j(\mathbf{r})}{\varepsilon}$$

The charge distribution of a single ion concentration is essentially the density of the ionic positions. It would be tedious and unfeasible to count every ion's position exactly as we are assuming too many things. Rather, one can use a statistical ensemble of ions via a cannonical ensemble, which is a good approximation here.

$$\rho_i(\mathbf{r}) = C_i e_0 Z_i \exp\left[-\beta Z_i e_0 \Psi(r)\right] \tag{5.2}$$

Here the energy of the system is defined as the potential energy scaled by the charge defined as $U = qV$. The full equation is known as the Poisson-Boltzmann equation, where $\Psi(\mathbf{r})$ is the potential for all space.

$$\nabla^2\Psi(\mathbf{r}) = -\frac{1}{\varepsilon_0\varepsilon_r}\sum_{i=1}^{\infty} C_i e_0 Z_i \exp\left[-\beta Z_i e_0 \Psi(r)\right] \tag{5.3}$$

There is a nice approximation we can make given that the potential of the solution is small. We linearize the exponential to first order here.

$$\begin{aligned}\nabla^2\Psi(\mathbf{r}) &\approx -\frac{1}{\varepsilon_0\varepsilon_r}\sum_{i=1}^{\infty} C_i e_0 Z_i \left[1 - \beta Z_i e_0 \Psi(r)\right] \\ &= -\frac{1}{\varepsilon_0\varepsilon_r}\sum_{i=1}^{\infty} \left[C_i e_0 Z_i - \beta C_i Z_i^2 e_0^2 \Psi(r)\right] \\ &= -\sum_{i=1}^{\infty}\left[\frac{C_i e_0 Z_i}{\varepsilon_0\varepsilon_r} - \kappa^2\Psi(r)\right]\end{aligned}$$

Where κ_i^2 is related to the debye-length, a unit of the electrostatic screening caused by the ionic interactions. The Debye length is defined as $\lambda_D = \kappa^{-1}$.

$$\kappa_i^2 = \frac{\beta C_i Z_i^2 e_0^2}{\varepsilon_0\varepsilon_r} \tag{5.4}$$

5.1.2 Monovalent Ionic Mixture

Suppose there is a single type of ion in the mixture. The sum collapses into a single term and the constant term is dropped.

$$\nabla^2\Psi(\mathbf{r}) = \kappa^2\Psi(\mathbf{r}) \tag{5.5}$$

This equation has an obvious solution which are the real exponentials;

$$\Psi(\mathbf{r}) = Ae^{-\kappa\mathbf{r}} + Be^{\kappa\mathbf{r}} \tag{5.6}$$

Chapter 6

Optics

Problem 6.1. *Solution Provided by Martin Jasinski, Neil Jadav.*

Solution. **(a)** The input beam is a single wavelength of light at a fixed power. $I_{in} = |\vec{E}_0| = 1$ W/cm^2. The input beam will hit a beam splitter and split in half towards two mirrors. Each mirror has a respective reflectivity, which reduces its intensity. The light then passes back towards the beam splitter, splitting one more time before reaching the detector. We can first write down the all the waves of interest. First, note that $I = |\vec{E}|^2$ and that $r = \sqrt{R}$

$$I_{in} = |E_0 \cdot \hat{x}| = 1 \quad [\mathbf{W/cm^2}] \tag{6.1}$$
$$I_{out} = |E_1 + E_2|^2 \tag{6.2}$$
$$= \left| E_0 \frac{r}{2} e^{i2kd_1} + E_0 \frac{1}{2} e^{i2kd_1} \right|^2 \tag{6.3}$$
$$= \left| E_0 \frac{0.9}{2} e^{i2kd_1} + E_0 \frac{1}{2} e^{i2kd_1} \right|^2 \tag{6.4}$$
$$\tag{6.5}$$

[2 Intensity Form]

Knowing the form of the output intensity, we can write out the norm of the out intensity.

$$\begin{aligned} I_{out} &= |E_0|^2 \left(0.45^2 + 0.5^2 + 0.45\cos(2k(d_1 - d_2))\right) \\ &= |E_0|^2 \left(0.4525 + 0.45\cos(2k\Delta d)\right)) \end{aligned}$$

[4 cosine Term]

Thus, we can see a minimum and a maximum based on the form of the cosine term. The minimum of cosine is -1 and the maximum is 1. In addition, we know from I_{in} that the total power $E_0^2 = 1$, so we can plug in the limits and obtain the min and max.

$$\boxed{\begin{aligned} I_{\mathbf{max}} &= 0.9025 \quad [\mathbf{W/cm^2}] \\ I_{\mathbf{min}} &= 0.0025 \quad [\mathbf{W/cm^2}] \end{aligned}} \tag{6.6}$$

[1 Answer]

[1 Units]

. . . . . .

(b) In the case of a second frequency aligning with the maximums at $\Delta d = 3\mu$m, we can inspect the cosine term and discover if there is any thing similar.

$$\begin{aligned} \cos(2k\Delta d) &= \cos\left(\frac{4\pi}{\lambda}\Delta d\right) \\ &= \cos\left(2\pi n\right), \quad n \in \mathbb{Z} \end{aligned}$$

[3 argument integer periodic]

Thus, we can solve the relation for λ_1;

$$n = \frac{2}{\lambda}\Delta d = \frac{6 \cdot 10^{-6}}{600 \cdot 10^{-9}} = \frac{1}{10^{-1}} = 10 \tag{6.7}$$

Neighboring frequencies with the **same** maxima at $\Delta d = 3 \cdot 10^{-6}$ can be found by varying n by ± 1;

$$\lambda_{n=11} = \frac{6 \cdot 10^{-}6}{11} = 545.\bar{45} \cdot 10^{-9} \tag{6.8}$$

$$\lambda_{n=9} = \frac{6 \cdot 10^{-}6}{9} = 666.\bar{6} \cdot 10^{-9} \tag{6.9}$$

[3 calculate nearby lambdas]

Thus, the closes wavelength from the original, ie

$$\arg\min_{\lambda_2}[\lambda_1 - \lambda_2]$$

$$\boxed{\lambda_2 = 545.\bar{45} \quad \textbf{[nm]}} \tag{6.10}$$

[1 Correct Answer]

. . . . . .

(c) Suppose $\lambda_1 \to \lambda_2$, what changes? We can see that the only wavelength dependent term is the argument of the cosine. This term changes the amount of Δd required to see the maxima/minima. This does **not** change the amount of signal that the detector measures. Thus, all answers except $\mathcal{A}$ are incorrect.

$$\boxed{\textbf{Correct Answer: } \mathcal{A}} \tag{6.11}$$

[5 Correct Answer]

Chapter 7

Digital Image Theory and Application

I have been working on images at my lab for quite some time now. This is a summary and documentation of the things i have learned while on the job.

7.1 Images as a Hilbert Space

At its core, images are taken by a camera array, which contains *pixels*. Each pixel is one point of the image, or a binning of millions of photons hitting the image, creating a smooth description of some object being photographed. Definitions of the image can be done in 2 ways; theoretical and practical.

Definition 7.1. *Let $f(x, y, z)$ be a field in $\mathbb{R}^3$. Let $\hat{P}$ be the projection operator such that $\hat{P} : \mathbb{R}^3 \to L_2(\mathbb{R}^2)$. Then the digital image is defined as $g(x, y) = \hat{P}f(x, y, z)$.*

the results of this defintion imply that the dimension z is integrated out. For instance, images of physical objects are the integration of the density function of space, $\rho(x, y, z)$, or when using a light emission system, the optical field density $I(x, y, z)$.

This defintion of the image lives in the vector-space of $\mathbb{R}^2$, which allows for a set of known transformations. Affine transformations are defined by a transformation matrix $\mathbf{T}$ and a shift $\vec{v}$.

$$f_T(\vec{x}) = f(\mathbf{T}^{-1}(\vec{x}) + \vec{b}) \tag{7.1}$$

For group transformations, we can define the action $(g \circ f)(\vec{x}) = f(g^{-1}\vec{x})$. This definition implies linearity for sums of scalar transformations of images;

$$g(af_1 + bf_2) = a(g \circ f_1) + b(g \circ f_2) \tag{7.2}$$

Furthermore, we can define the group as being part of $SO(2)$. This group includes the rotation matrices, which gives the norm-preservation property;

$$|g \circ f| = |f|_{L_2} \tag{7.3}$$

Naturally, the functions we produce want to have some sense of similarity between them. Ie, how similar are two images? To do this, we can re-use the idea of a generalized angle. Two images can be the same in terms of angle, but have different magnitudes. This can be interpretted as a measure in distance. From QM, we know that a square-integratable set of functions with angle and distance are defined in the Hilbert space.

Definition 7.2. *Let $\mathscr{H}$ denote the Hilbert space; Elements of $\mathscr{H}$ must satisfy the following criteria. Let $a, b \in \mathscr{H}$:*

1. *Elements have an inner product:* $\langle a, b\rangle$
2. *Norm Property:* $\langle a, a\rangle > 0$
3. *Zero Property:* $\langle a, a\rangle = 0$ *if* $x = 0$
4. *Inner product commutation:* $\langle a, b\rangle = \langle b, a\rangle^*$
5. *Linearity for constants:* $\langle a, e_1 b_1 + e_2 b_2\rangle = e_1^* \langle a, b_1\rangle + e_2^* \langle a, b_2\rangle$
6. *Angle:* $A(a, b) = \langle a, b\rangle$
7. *Distance:* $d(a, b) = \sqrt{\langle a - b, a - b\rangle}$

The inner product of two functions is evaluated as their integral over the domain, hence;

$$\langle f(x, y), g(x, y)\rangle = \iint f g^* dx dy \tag{7.4}$$

From QM, we know about the existence of unitary operators. Below is a defintion that we can use to define rotation operations.

Definition 7.3. *Let U be a unitary operator acting on f. $U_\theta : L_2(\mathbb{R}^2) \to L_2(\mathbb{R}^2)$ such that the norm is preserved $|U_\theta f| = |f|$.*

7.1.1 Rotational Symmetry

Let U_θ be a unitary rotation operator in $SO(2)$.

We can define exact symmetry as an identical copy due to rotations; $U_\theta f = f$. We know that our images will contain some noise; $U_\theta(f + \eta) = f + U_\theta \eta$. These disturbances will not be symmetrical.

Definition 7.4. *Let the rotational symmetry be defined as how similar the image is to its rotated counterparts normalized by the norm of the image.*

$$S(f) = \sup_\theta \frac{\langle f, U_\theta f\rangle}{|f|^2} = 1 - \inf_\theta \frac{|f - U_\theta f|}{|f|} \tag{7.5}$$

We can also define the maximum symmetry being $S(f) = 1$, whereas minimum symmetry, or a total orthogonal set, $S(f) = 0$.

This function is bounded on $[0, 1]$. The supremum clause defines the largest possible deviations from a perfect symmetry.

This definition uses the auto-correlation of the image with its rotational counter-part, however, integrates over the radial-ness of the image. For parts of the image that are only radially symetric for a set of angles, we can define the measure over a radial average;

For instance, say that the image is symmetric over a wide range of angles, but contains portions where the symmetry is broken by a mis-alignment or blip, these would be integrated over in the symmetry measure.

Definition 7.5. *Let the radial average be defined as the integral over angles θ_0.*

$$f_0(r, \Delta\theta, \theta_0) = \frac{1}{2\pi} \int_{\theta_0}^{\theta_0 + \Delta\theta} f(r, \theta) d\theta \tag{7.6}$$

Then, define the symmetry metric as the ratio between the norm of the image, and the norm of the radial averaged-portion;

$$S(r, \theta_0, \Delta\theta) = \frac{|f_0|^2}{|f|^2} \tag{7.7}$$

Chapter 8

Computer Science and Algortihms

This is mainly about algorithmns as i dont know shit about computers

8.1 Github

Code is writing. Similar to how books are shared in a libary, code is shared through a worldwide network known as Github. Github consists of repositories, which act as book covers, which ensure that every project has a self-contained way to share it, read it, and use it.

This section describes Github and some of the features that I have found useful as a graduate student.

8.1.1 repositories and how to use them

Creating a new repository is as simple as clicking "New repository," opening it on the computer, and "Push"-ing it onto the github network.

Whenever you are adding something new to a github repository, it is wise to open a "Branch." Branches are designed to copy the current version of the code without breaking the current working version, typically referred to as "main" or "master." Branches can be made by clicking on the branch icon and creating a new Branch.

When you want to collaborate on a piece of code, or specifically, want to change something, one can submit a "Pull Request" in which collaborators can offer advise to changes on code before they are pulled.

To merge branches, one can submit a merge request. Once the merge request has been passed, one can delete a branch once it is transferred onto the main branch.

8.2 Working in Python

Python is an easy scripting language with many perks in the scientific community. Below are some things that i picked up while using it

8.2.1 numpy

Best library there is.

8.2.2 matplotlib

second ebst library there is/

8.2.3 dask

In many scientific and engineering applications, the amounts of data which need to be analyzed range from single text files, to large 6Tb datasets of images, metadata, and many involved parameters. In large cases, datasets cannot be opened all at once; for instance, how can a 5Tb dataset be parsed into the active RAM memory if even beefy systems have up to 128 gB of ram?

To parse memory of this size, many approaches can be taken, as shown in the C++ section of this chapter. Luckily, developers more experienced than myself have written many APIs to do so. one such APi is known as DASK, a PYTHON library for parallel and distributed computing.

The dask API consists of three main functionalities, **tasks**, **dataframes**, and **arrays**.

dask.tasks

dask.dataframes

dask.array

The DASK array, referred in this section as simply a darray, consists of a coordinate array of many NUMPY arrays.

To learn this system, lets use an example in the lab - the Nikon .ND2 format. The .ND2 format is an image format which stores image data, as well as many meta-data, together in a single container which can be utilized in downstream pipelines. This format is also greatly used in many biological applications. The main reason to switch this format into a DASK array is to allow to offload the information of larger .ND2 files into chunks, allowing for faster analysis.

Below is an outline of the .ND2 structure

```
video.nd2
├── Coordinates
│   ├── T
│   │   ├── values
│   │   └── attributes (units)
│   ├── C
│   │   └── values (color as a string)
│   ├── Z
│   │   ├── values
│   │   └── attributes (units)
│   ├── Y
│   │   ├── values
│   │   └── attributes (units)
│   └── X
│       ├── values
│       ├── attributes (units)
│       ├── values
│       └── attributes (units)
└── Attributes
    ├── metadata:
    └── processed:
```

Using the TOOLSJASINSKI code below, we can initialize this as a dask array;

```
from src.toolsjasinski._parse_video import parse_video

path = r'src\toolsjasinski\tests\ALEX_Nanowells_10mWRL_50mWGL_3MM_2min.nd2'

video = parse_video(path)
```

Now, the object VIDEO contains an xarray.Dataarray consisting of the chunked video information, as well as an attributes dictionary. We can see this by printing some properties of the file in

question;

```
print(video)
print(f"File Size: {video.nbytes/1e9:.2f} Gb")
print(video.sizes)

> OUTPUT

<xarray.DataArray '_dask_block-aacbdb2a18dff2e73fd9cf0b459a7cc1' (T: 2378,
                                                                  C: 2, Z: 1,
                                                                  Y: 512, X: 512)> Size: 2GB
dask.array<transpose, shape=(2378, 2, 1, 512, 512), dtype=uint16, chunksize=(1, 2, 1, 512, 512
Coordinates:
* T        (T) float64 19kB 0.0 50.48 101.0 ... 1.199e+05 1.199e+05 1.2e+05
* C        (C) <U5 40B 'red' 'green'
* Z        (Z) float64 8B 0.0
* Y        (Y) float64 4kB 0.0 0.16 0.32 0.48 0.64 ... 81.28 81.44 81.6 81.76
* X        (X) float64 4kB 0.0 0.16 0.32 0.48 0.64 ... 81.28 81.44 81.6 81.76
Attributes:
    metadata:   {'metadata': Metadata(contents=Contents(channelCount=2, frame...
    processed:  images=[<1 field_type>] creator='N/A'
File Size: 2.49 Gb
Frozen({'T': 2378, 'C': 2, 'Z': 1, 'Y': 512, 'X': 512})
```

8.3 Working in C++

A great language to use in physics programming is C++ and Python. C++ is described first as it is more technical and requires a tad more effort.

8.3.1 Makefiles

Suppose that the file structure of a given project consists of a main file, and some function files that contain functions that are used in the main file. Such a file structure is shown as:

```
my-project/
├── iterate.cpp/
├── iterate.h/
├── finalize.cpp/
├── execute.cpp/
├── initialize.cpp/
└── wait.cpp /
```

This complex file structure requires linking of each header file, known as a library, to the main file which is labelled as **iterate.cpp**. This can be done using a program known as Makefile. Makefile is a library manager that links library paths together during the compilation step. Below is the Makefile for the above example:

```
CC = clang++

all:
    $(CC) iterate.cpp initialize.cpp execute.cpp finalize.cpp wait.cpp -o one
```

Let's walk through this. **CC** is a variable for the language. Here, we use the CLANG++ compiler. Other options are G++ or GCC. In Makefile, the variable is always a string and can be initiated by setting a name and a value. To call the variable, use the $ symbol before the variable.

```
CXX = g++
CXXFLAGS := -Wall
CXXFLAGS +=  -O2
```

```
main:
 $(CC) $(CXXFLAGS)iterate.cpp execute.cpp finalize.cpp initialize.cpp wait.cpp -o one

// called g++ -Wall -02 iterate.cpp execute.cpp finalize.cpp initialize.cpp wait.cpp -o one
```

8.3.2 CMake

As nice as Makefiles are, it appears that they ar enot used that often anymore in larger projects. Why is this? probably due to power-scaling of these computer science people - whatever, not my problem.

To use CMake, we need to first download it.

A CMake project consists of a file and a CMakeLists.txt file. The CMakeLists.txt file is a specific file, similar to a Makefile. Below is an example file structure, as well as an associated CMakeLists.txt file along with some comments.

```
my-project/
├── app/
│   └── main.cpp/
├── include/
│   ├── json/
│   │   └── json.hpp/
│   └── iscat_file_load/
│       └── iscat_file_load.hpp/
├── src/
│   └── iscat_file_load.cpp/
└── CMakeLists.txt/
```

```
cmake_minimum_required(VERSION 3.20)

project(some_project CXX)

#json
add_library(jsonLib INTERFACE)
target_include_directories(jsonLib INTERFACE ${CMAKE_CURRENT_SOURCE_DIR}/json)

#iscat file load
add_library(iscat_file_load src/iscatloader.cpp)
target_include_directories(iscat_file_load PUBLIC include)
target_link_libraries(iscat_file_load PUBLIC HDF5::HDF5)

#HDF5
#add_subdirectory(include/h5pp)
find_package(HDF5 REQUIRED COMPONENTS CXX)

# add executable
add_executable(main ${PROJECT_SOURCE_DIR}/app/main.cpp)
# link libraries
target_link_libraries(main PRIVATE jsonLib)
target_link_libraries(main PRIVATE iscat_file_load)
```

The explanations of each flags are listed in the list below;

- cmake_minimum_required - Minimum required version of cmake
- project - defined project variable and version of C++
- add_library - adds library. See explanation

- target_include_directories - tells the compiler where **.h** and **.hpp** files are located.
- target_link_library - The compiler will provide the target with the library. For instance passing in **(main_program PRIVATE jsonLib)** will provide main with the functions and definitions in the library **jsonLib**. The flag ensures that nothing derived from the target main_program will inherit functions in jsonLib.
- find_package - searches for external libraries.

8.3.3 C++ Library Documentation

This section describes documentation for certain libraries as I use them so that i can understand what they do. They shouldn't be used as a replacement for the actual documentation, just notes.

8.3.4 HDF5 Files and their API

The hierarchical data format, HDF, is a set of file formats that is used to store and organize large amounts of data. These are developed for supercomputer purposes, as well as general large datasets, such as those used in machine learning in recent years.

The HDF5 format consists of a large multi-dimensional array of identically typed elements. The datatype is defined across the dataset. The multi-dimensional array consists of a **Rank** and a set of **Dimensions**. The rank is a simple extensions from matrix algebra as the rank of a matrix.

The main trouble in HDF5 files is their nested-ness, specifically in the case of datasets where there are long nested strcutures. In this case, it is often not obvious how to get the data that is stored. For example, we can use a dataset type that has been produced after imaging some particles. This is a particle-tracking dataset coined as the **ROI_Traces** dataset style:

8.4 Dynamic Programming

Dynamic programming is something i dont understnad

8.4.1 2D Chain-Link Algorithm

This is text

Algorithm 1 Dynamic Programming for Best 2D Chain of Points

Require: min_idx $= \{\text{min_idx}[t]\}_{t=0}^{F-1}$, list of candidate positions per frame
min_vals $= \{\text{min_vals}[t]\}_{t=0}^{F-1}$, list of candidate values per frame
$w = (w_x, w_y)$, weighting for distance computation (default $(1, 1)$)
max_step, optional maximum allowed step distance
Ensure: idx_chain, x_chain, y_chain, the indices, positions, and values of the optimal path
Initialize $F \leftarrow |\text{min_idx}|$
Initialize $cost[i] \leftarrow 0$ for all i in frame 0
Initialize $prev \leftarrow []$ ▷ to store predecessor indices
for $t = 1$ to $F - 1$ **do**
    $A_x \leftarrow \text{min_idx}[t-1]$, $B_x \leftarrow \text{min_idx}[t]$
    $A_y \leftarrow \text{min_vals}[t-1]$, $B_y \leftarrow \text{min_vals}[t]$
    Compute pairwise differences:

$$\Delta_x[i,j] = |A_x[i] - B_x[j]|, \quad \Delta_y[i,j] = |A_y[i] - B_y[j]|$$

    Compute weighted Euclidean distances:

$$D[i,j] = \sqrt{(w_x \Delta_x[i,j])^2 + (w_y \Delta_y[i,j])^2}$$

    if max_step is not None **then**
        Set $D[i,j] \leftarrow \infty$ if $D[i,j] > \text{max_step}$
        if no finite $D[i,j]$ exists **then**
            return empty path
        end if
    end if
    Compute cumulative path cost matrix:

$$M[i,j] = cost[i] + D[i,j]$$

    For each node j in frame t, find predecessor $p[j]$ minimizing $M[:, j]$:

$$p[j] = \arg\min_i M[i,j], \quad new_cost[j] = M[p[j], j]$$

    Update $cost \leftarrow new_cost$
    Append p to $prev$
end for
Initialize $j \leftarrow \arg\min_i cost[i]$ ▷ index of minimal total cost in last frame
Initialize idx_chain $\leftarrow [0]^F$, set idx_chain$[F-1] \leftarrow j$
for $t = F - 2$ down to 0 **do**
    idx_chain$[t] \leftarrow prev[t][\text{idx_chain}[t+1]]$
end for
Compute chains:

$$\text{x_chain}[t] = \text{min_idx}[t][\text{idx_chain}[t]], \quad \text{y_chain}[t] = \text{min_vals}[t][\text{idx_chain}[t]]$$

return idx_chain, x_chain, y_chain

Chapter 9

Machine Shop Fundamentals

This section covers the basics that I have learned while taking a course on machining. The basics include usage of various instruments, such as a band-saw, drill press, and lathe.

The basics of drawings, screws, and conventions styles are all listed here as they go.

9.1 Safety

Safety is the most important thing in the machine shop and lab.

That being said, here are a list of safety rules that if one adheres to, they should never be injured.

1. Wear safety goggles at all times. When using moving machines, don't use gloves unless you are aware that if they are caught in something, you will be dragged into that machine.
2. Be aware of yourself and your surroundings.

The machine shop has a lot of moving parts. If you are aware of what those parts are and know not to displace or cause those parts to move near you, this will improve your odds of survival.

3. No hanging clothing or long hair. Ensure jewelry is removed before operation of moving parts. Hair should be tied up or kept in a hat.
4. When operating a device that is spinning, remember, as with gloves, that if something is caught in it, it will be dragged into the rotating machine until it is stopped.

Recall from Chap. TBD that objects with angular momentum will conserve that momentum unless acted upon. This principle is known as Newton's second law, which is often forgotten about unless stated explicitly.

5. Never use your hands to remove chips as they can be sharp.
6. Keep areas clean, as they should be. If something is out of place, put it back.

This cleanliness principle is actually very useful in life as well. The area you reside in will largely dictate your mood and performance. Removing tripping hazards is also a bonus.

7. Never work alone.
8. Don't use a cell-phone or watch TV. The machine shop is a place of work, not play.
9. If a machine is broken or not working properly - report it. Peoples safety is more important

than a broken piece of replaceable[1] equipment.

10. If you are hurt, please get help before you no longer can[2].

9.2 Units

There are several methods of units in the machine shop. The two main ways of describing length are the imperial and metric measures. The main dimensions which need to be described are units of length, depth, radius or diameter, screw type, gear type, and angles and bores. The dimensions and tolerances, set by the American Society of Mechanical Engineers (ASME) is a guideline for a universal set of standards for technical drawings.

9.2.1 Length

Length, denoted by an arrow symbol, describes the length of dimensions of a drawing.

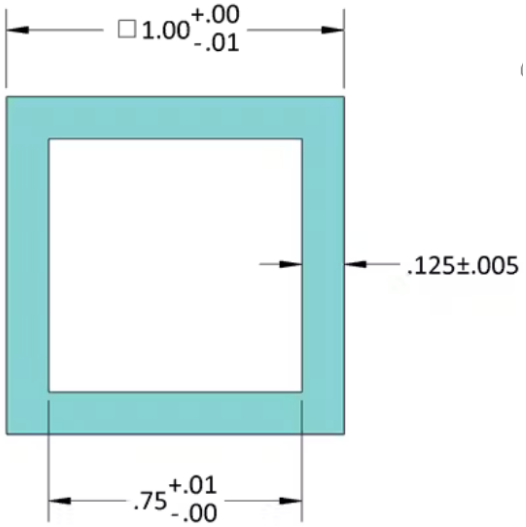


Figure 9.1: Basic Dimensions with size tolerances of a box shape. Lengths, denoted by a pair of arrows, are listed alongside a numerical measurement, typically associated with a unit convention such as imperial or metric. The tolerance, denoted by the plus/minus sign, writes the maximum and minimum lengths of the part.

9.2.2 Diameter

The unit of diameter, typically denoted by a ∅ or **DIA**, is associated with the length of the end to end, passing through the center of a circular shape or angle, shown in the example of Fig. 9.2.

9.2.3 Screws and Holes

[1] Although, if you had to save the entirety of CERN or one person's life, this might raise some questions. Should I have a section on ethics?

[2] In the event of death of life-altering infection.

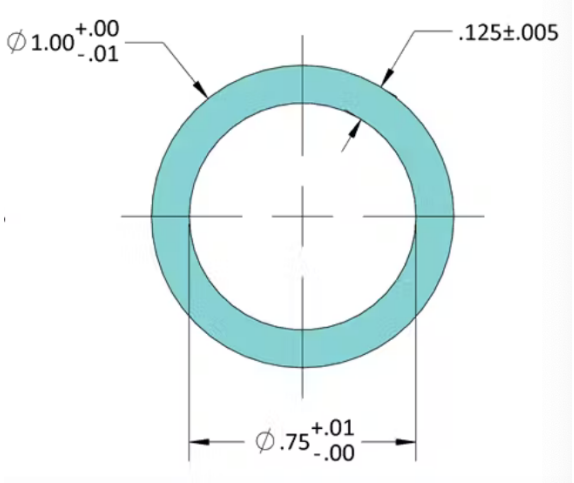


Figure 9.2: Basic Dimensions with size tolerances of a circular shape. Lengths, denoted by a pair of arrows, are listed alongside a numerical measurement, typically associated with a unit convention such as imperial or metric. The tolerance, denoted by the plus/minus sign, writes the maximum and minimum lengths of the part. The diameter of the circle is noted by the symbol ∅, which explicitly states the diameter of the shape. Here, pictured is the nominal diameter, or outer-diameter abbreviated OD. The ordinal diameter, or inner-diameter abbreviated ID.

9.3 Technical Drawings

Picture yourself in the machine shop. You just finished your CAD and model. You have the parts and the machines. How are you going to turn the block of metal into a finished product? The answer lies in a drawing that a machinist, like yourself or a friend, can easily make and digest. This document loosely following the NASA Engineering Drawing Standards Manual, found in Ref. [1].

Lines are the fundamental block of all drawings. Lines should be all roughly the same thickness/width and straight. Different styles of lines can be used to represent different things. Several lines, such as dimensions, centering or symmetries, hidden areas, or breaks can be represented by a set of dotted or partially dashed lines, shown in Fig. 9.3.

Phantom lines can be used to indicate an alternate position of the part or a repeated detail. For instance, consider a spring with a repeated coil structure. Here, phantom lines are used to show a symmetry along various angles or variations.

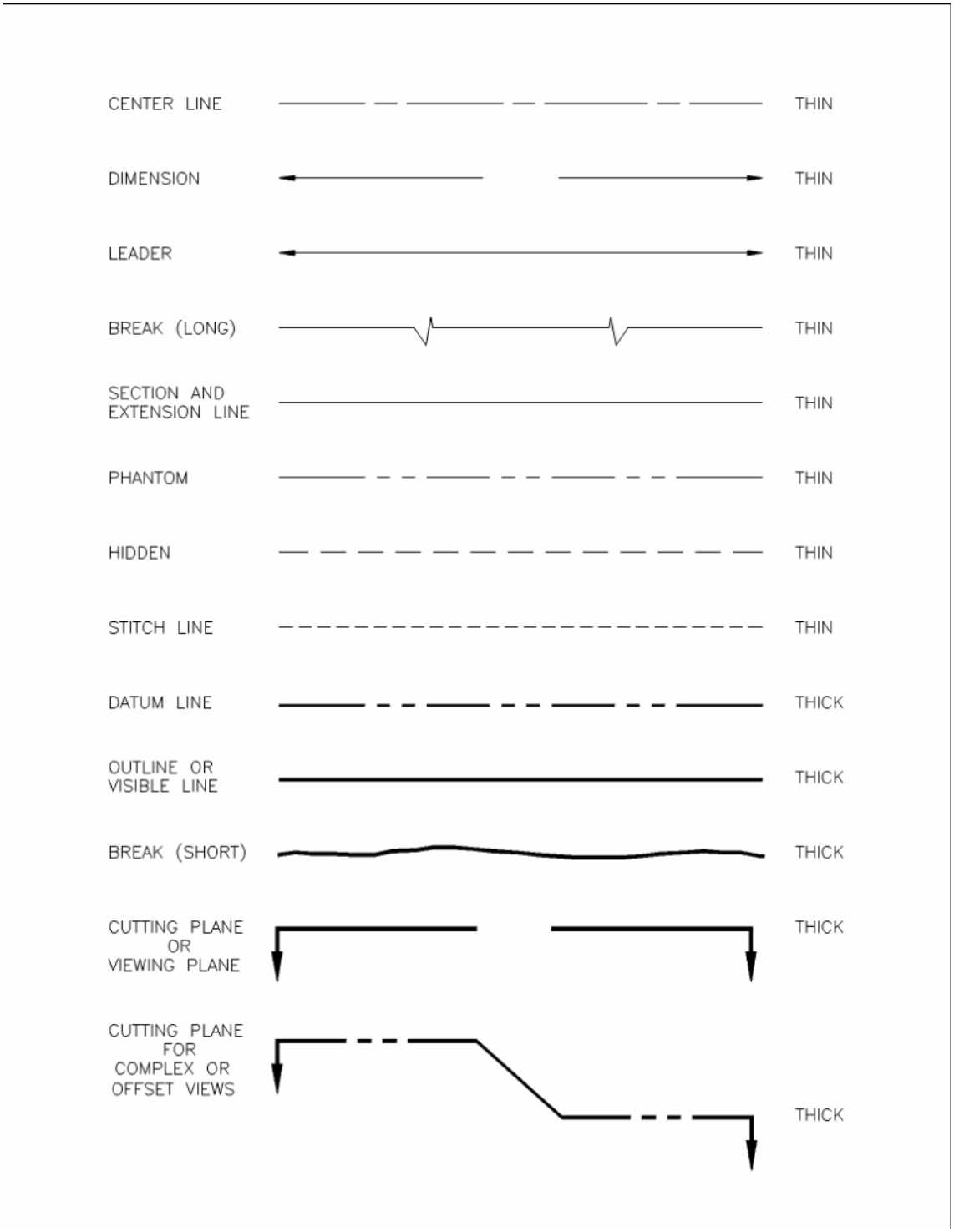


Figure 9.3: Line Types and Conventions.

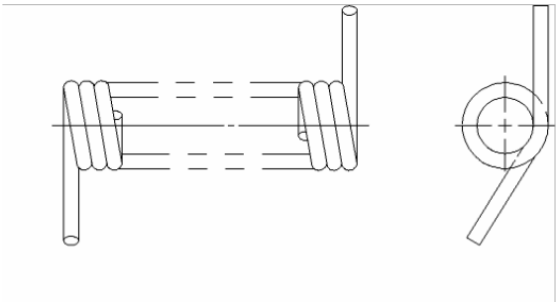


Figure 9.4: Coiled spring with repeated details.

9.3.1 Projection Types

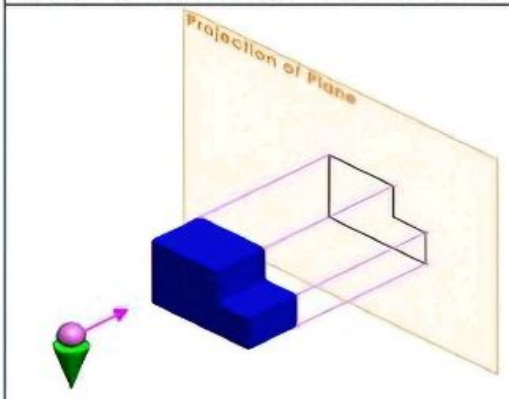
The projection of a drawing is important to consider. For instance, the projection of a cut can dictate the topology of the structure as one must consider which face details are made on.

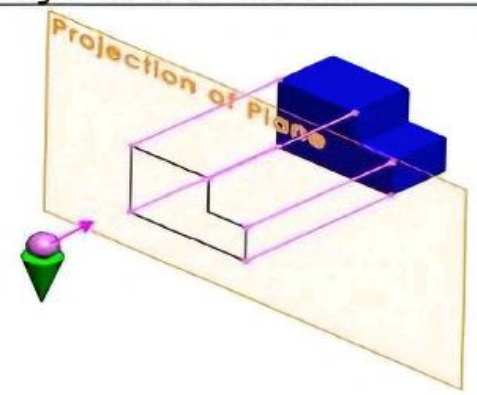
The two main projection methods are known as the first-angle and third-angle projection, which are typically found in the ISO and ASME respectfully. The ISO is used in most countries, while the ASME is used in the US, Canada, Japan, and Australia.

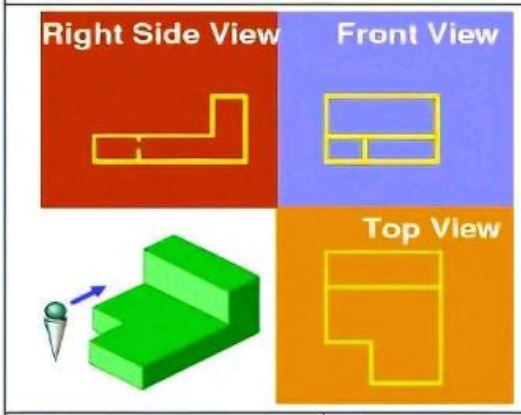
9.3.2 Examples

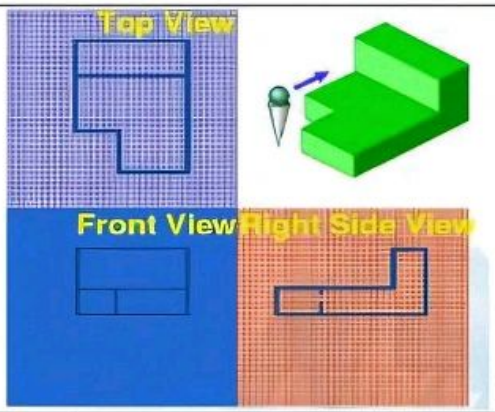
Example technical drawings, as well as a template for hand-drawn drawings can be found in Fig. 9.6, as well as an attached PDF.

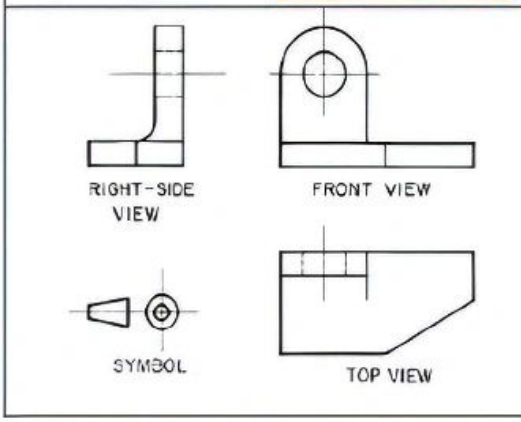
First Angle Projection	Third Angle Projection
The object is imagined to be in first quadrant.	The object is imagined to be in third quadrant.
The object lies between the observer and the plane of projection.	The plane of projection lies between the observer and the object.
The plane of projection is assumed to be non transparent.	The plane of projection is assumed to be transparent.
When views are drawn in their relative positions the Top view comes below the Front view, Right side view is drawn to the left side of the elevation.	When views are drawn in their relative positions the Top view comes above the Front view, Right side view is drawn to the right side of the elevation.











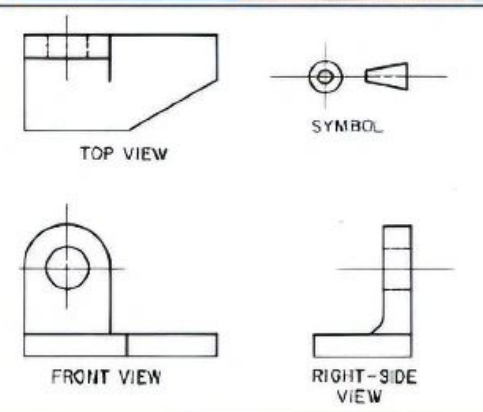


Figure 9.5: First and third angle projection style.

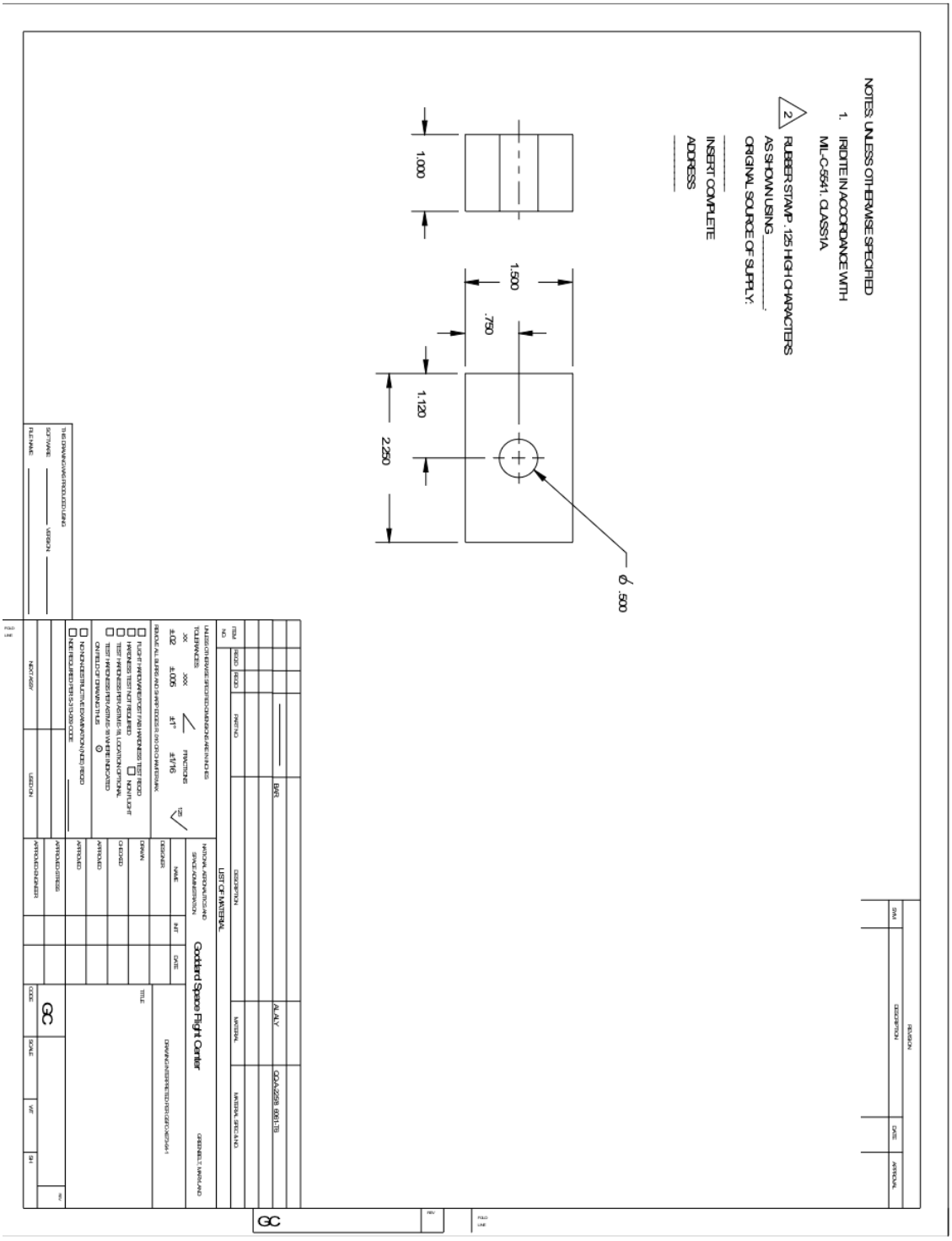
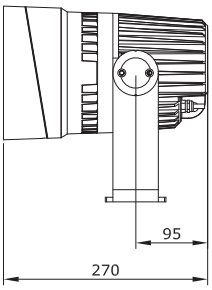
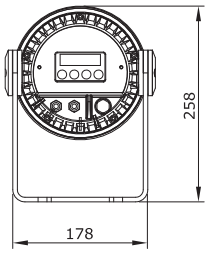
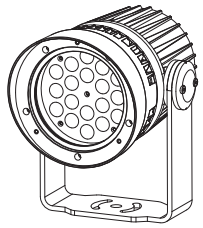


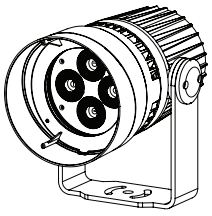
Figure 9.6: Example technical drawing.



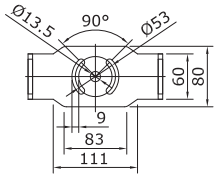


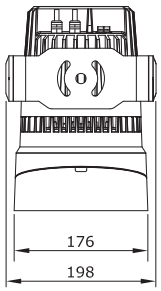


Exterior Wash 100/120
without snoot



Exterior Wash 110
with snoot





Rev.	Date	Resp.	CO no. / Comment
A	23.08.2018	rd_lto	First release

Martin	First used in		Material
Item to be			General linear tolerance
Item to be			General angular tolerance
Item to be		Color	Format A3 / Scale 1:5 / Unit mm
Description: Exterior Wash 100 Series with snoot			Item no.-Rev.

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Martin™ General Technical Specification apply to this item

9.4 Band-saw

9.5 Drill-press

9.6 Lathe

Bibliography

[1] National Aeronautics and Space Administration. *NASA Systems Engineering Handbook*. Special Publication NASA/SP-2016-6105. Version Revision 2. Washington, D.C.: National Aeronautics and Space Administration, 2017. URL: https://ntrs.nasa.gov/citations/20170001761.

[2] Maxwell Rosenlicht. *Introduction to Real Analysis*. Ed. by 1. Important Books, 2013.